PINN_M3DC1

Module 1

Reduced Resistive-MHD Physics and Conventions

A first-principles, textbook-style introduction for readers new to plasma physics

Project model

Two-dimensional, incompressible, two-field reduced resistive magnetohydrodynamics on a doubly periodic Cartesian domain

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Physics note for the synthetic-data and physics-informed neural-network proof of concept

Preface: what this note is designed to do

This note is deliberately written from the beginning. It does not assume a prior course in plasma physics or magnetohydrodynamics. It does assume that the reader is comfortable with multivariable calculus, ordinary differential equations, and elementary classical mechanics and electromagnetism. Every project variable is defined before it is used in the final model.

The aim is not merely to display two partial differential equations. The aim is to make those equations physically intelligible: where they come from, what each term does, which assumptions make them possible, how the signs are fixed, what is conserved, what is dissipated, how magnetic reconnection can occur, and how the equations become constraints in a physics-informed neural network.

The initials **MHD** mean *magnetohydrodynamics*: the fluid theory of an electrically conducting medium interacting with a magnetic field. The initials **PINN** mean *physics-informed neural network*: a neural network trained using physical equations in addition to, or instead of, measured or simulated data. The abbreviations **2D** and **3D** mean two-dimensional and three-dimensional. The code name **M3D-C1** denotes a nonlinear extended-MHD code developed for magnetized-plasma calculations. In that name, C1 refers to C^1 finite elements, whose represented fields and first spatial derivatives are continuous across element boundaries [3, 4].

Important distinction

This project does *not* run M3D-C1 and does *not* train on genuine M3D-C1 output in its first stage. It numerically generates synthetic trajectories with an independent, much simpler two-dimensional reduced-MHD model. The result can demonstrate a surrogate-learning workflow shaped for later M3D-C1 data, but it is not a validated M3D-C1 surrogate.

How to read the note

The conceptual dependency is

$$\begin{aligned}\text{plasma as a fluid} &\longrightarrow \text{one-fluid resistive MHD} \longrightarrow \text{two-dimensional reduction}\\ &\longrightarrow \text{flux and stream functions} \longrightarrow \text{two-field equations}\\ &\longrightarrow \text{conservation and reconnection} \longrightarrow \text{PINN equation residuals.}\end{aligned}$$

If a later section feels abrupt, the safest response is to return to the immediately preceding link in this chain. Sections 2–5 establish the background. Sections 6–12 derive the project equations. Sections 11–19 freeze the computational conventions. Sections 15–17 explain the physics and the relationship to the larger project.

Learning objectives

After working through this module, the reader should be able to:

1. explain why MHD is a fluid model even though plasma physics can also be kinetic;
2. state the one-fluid resistive-MHD equations and identify their unknown fields;
3. distinguish electrical resistivity, magnetic diffusivity, and the dimensionless project coefficient η;
4. derive divergence-free magnetic and velocity fields from scalar potentials ψ and U;
5. derive $J = -\nabla^2\psi$ and $\omega = \nabla^2 U$ from the frozen sign conventions;
6. explain the Poisson bracket $[f, g]$ as two-dimensional advection;
7. derive and interpret both two-field evolution equations;
8. nondimensionalize the model using the Alfvén speed and Alfvén time;
9. derive the total-energy dissipation law and use it as a verification test;
10. identify magnetic O-points, X-points, islands, current sheets, and reconnection;
11. write the exact PINN residuals without changing a sign or normalization; and
12. explain precisely what the reduced model retains and omits relative to M3D-C1.

Contents

1 Scope, model hierarchy, and scientific claim

1.1 Why begin with a reduced model?

A fusion plasma can contain many coupled processes: bulk fluid motion, magnetic-field evolution, pressure and temperature transport, electron-ion differences, fast particles, radiation, atomic physics, interaction with conducting structures, and complex three-dimensional geometry. A production code must decide which of these processes to represent and at what fidelity. A first surrogate-learning experiment should not begin by reproducing every process simultaneously. It should begin with a model small enough that its equations, numerical solution, and neural residuals can all be verified independently.

The selected model has two scalar state fields:

- $\psi(x, y, t)$, the dimensionless magnetic-flux function; and
- $U(x, y, t)$, the dimensionless flow stream function.

From them we derive

- $J(x, y, t)$, the dimensionless out-of-plane current variable; and
- $\omega(x, y, t)$, the dimensionless out-of-plane fluid vorticity.

The independent variables are Cartesian coordinates x and y, and time t. The initial parameter vector is

$$\boldsymbol{\mu} = (\eta, \nu, A_0), \tag{1.1}$$

where η is dimensionless magnetic diffusivity, ν is dimensionless kinematic viscosity, and A_0 is the dimensionless amplitude of the initial perturbation. The exact initial-condition formula involving A_0 belongs to Module 2.

1.2 A hierarchy of descriptions

The word *reduced* means that a controlled set of variables or physical effects has been removed from a more general model. It does not mean "unphysical." It means "valid only under a stated set of assumptions."

1.3 The three claims that must remain separate

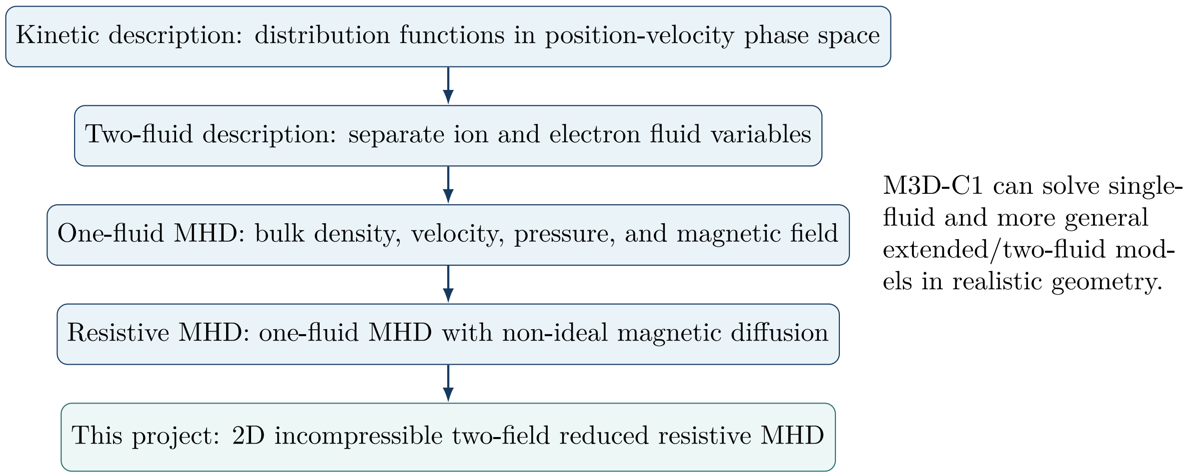

Figure 1: A conceptual hierarchy. Each downward step discards degrees of freedom under additional assumptions. M3D-C1 is not simply the box immediately above the project model; it spans a substantially broader set of models.

Claim	Scientifically defensible meaning
A	A parameterized PINN can be tested against independently generated synthetic trajectories of a stated two-field reduced resistive-MHD model.
B	The inputs, outputs, metadata, diagnostics, and train/test organization can illustrate how a later workflow could accept M3D-C1 exports.
C	A neural surrogate accurately predicts actual M3D-C1 results over a measured domain of equilibria, closures, meshes, perturbations, and time. This claim requires genuine M3D-C1 data and lies outside the initial project.

Module 1 provides the physical foundation for Claim A and the vocabulary needed to describe Claim B honestly. It provides no evidence for Claim C.

2 Mathematical language used throughout the module

This section defines the notation needed later. It is not a complete vector-calculus course; it is a compact dictionary between the mathematics and the physical statements.

2.1 Coordinates, scalar fields, vector fields, and unit vectors

We work first in three-dimensional Cartesian space with coordinates (x, y, z). The unit vectors $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, and $\hat{\boldsymbol{z}}$ have unit length and point along the positive coordinate axes. A *scalar field* assigns one number to every location and time; pressure $p(x, y, z, t)$ is an example. A *vector field* assigns a vector; velocity

$$\boldsymbol{v}(x,y,z,t) = v_x\widehat{\boldsymbol{x}} + v_y\widehat{\boldsymbol{y}} + v_z\widehat{\boldsymbol{z}} \tag{2.1}$$

is an example. Subscripts on a vector component, such as v_x, identify the component and do not denote differentiation.

The project later imposes $\partial/\partial z = 0$. This means that every evolved field is independent of z. The z direction is then called an *ignorable* or *out-of-plane* direction. A vector can still have a z component even when its fields do not vary with z.

2.2 Partial derivatives and the gradient

The symbol

$$\frac{\partial f}{\partial x} \equiv f_x \tag{2.2}$$

is the partial derivative of a scalar field f with respect to x, holding the other independent variables fixed. We will use both forms. The gradient collects the three spatial partial derivatives into a vector:

$$\nabla f = \widehat{\boldsymbol{x}}\frac{\partial f}{\partial x} + \widehat{\boldsymbol{y}}\frac{\partial f}{\partial y} + \widehat{\boldsymbol{z}}\frac{\partial f}{\partial z}. \tag{2.3}$$

The gradient points in the direction of fastest local increase of f.

When only x and y derivatives are present, we write

$$\nabla_\perp f = \widehat{\boldsymbol{x}} f_x + \widehat{\boldsymbol{y}} f_y. \tag{2.4}$$

The subscript $\perp$ anticipates that this plane is perpendicular to the constant guide field introduced later.

2.3 Divergence, curl, and Laplacian

The divergence of a vector field measures local expansion or compression:

$$\nabla\cdot\boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}. \tag{2.5}$$

A flow is *incompressible* when $\nabla\cdot\boldsymbol{v} = 0$.

The curl measures local rotation:

$$\nabla\times\boldsymbol{v} = \begin{vmatrix} \widehat{\boldsymbol{x}} & \widehat{\boldsymbol{y}} & \widehat{\boldsymbol{z}} \\ \partial_x & \partial_y & \partial_z \\ v_x & v_y & v_z \end{vmatrix}. \tag{2.6}$$

The vorticity vector is defined as $\boldsymbol{\omega} = \nabla\times\boldsymbol{v}$. In a two-dimensional in-plane flow it has only a z component.

The Laplacian of a scalar is the divergence of its gradient:

$$\nabla^2 f \equiv \nabla \cdot \nabla f = f_{xx} + f_{yy} + f_{zz}. \tag{2.7}$$

In two dimensions,

$$\nabla_\perp^2 f = f_{xx} + f_{yy}. \tag{2.8}$$

Repeated subscripts denote repeated differentiation: $f_{xx} = \partial^2 f/\partial x^2$.

2.4 Local time change and material time change

The partial derivative $\partial f/\partial t$ measures the change observed at one fixed spatial point. A moving fluid element experiences an additional change because it travels through spatial gradients. Its *material derivative* is

$$\frac{Df}{Dt} \equiv \frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla f. \tag{2.9}$$

The term $\boldsymbol{v} \cdot \nabla f$ is *advection*: transport of f by the bulk motion.

Definition

Advection moves a pattern with the flow. Diffusion smooths a pattern by reducing sharp gradients. A prototype advection-diffusion equation is

$$\frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla f = D\nabla^2 f,$$

where D is a diffusivity with dimensions of length squared per time.

2.5 Useful identities

The derivations use the identities

$$\nabla \cdot (\nabla \times \boldsymbol{A}) = 0, \tag{2.10}$$

$$\nabla \times (\nabla\phi) = 0, \tag{2.11}$$

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}. \tag{2.12}$$

The first states that a field written as a curl is automatically divergence-free. The second states that a pure gradient has no curl. The third connects the curl of the current to the Laplacian of the magnetic field.

3 From particles to a conducting fluid

3.1 What is a plasma?

A plasma is an ionized medium containing mobile charged particles, most commonly positively charged ions and negatively charged electrons. It is not defined merely by being hot. Its defining behavior is collective: electromagnetic fields generated by many particles influence the motion of many other particles.

The *Debye length* is the characteristic distance over which a plasma screens an electrostatic disturbance. On length scales much larger than this screening length, most fusion plasmas are approximately *quasi-neutral*: the positive and negative charge densities nearly cancel. Quasi-neutrality does not mean that the electric field is zero. Very small departures from exact neutrality can support dynamically important electric fields.

3.2 Three common levels of description

Kinetic description. For each species s (for example, ions or electrons), a distribution function $f_s(\boldsymbol{x}, \boldsymbol{v}, t)$ gives density in a six-dimensional phase space formed by three position coordinates $\boldsymbol{x}$ and three velocity coordinates $\boldsymbol{v}$. This description retains velocity-space structure, particle resonances, non-thermal populations, and many collisionless effects.

Fluid description. Velocity moments of f_s produce fields in ordinary space. The zeroth moment gives number density, the first gives bulk velocity, and the second is related to pressure and temperature. A fluid model evolves a finite set of such moments and requires a *closure*: an additional physical relation that approximates unevolved higher moments using the retained variables.

Magnetohydrodynamic description. MHD combines the charged species into a macroscopic conducting fluid and couples its mass, momentum, and thermodynamic evolution to reduced Maxwell equations. It is therefore a fluid theory, even though its coefficients and closures ultimately arise from particle physics.

3.3 Two-fluid and one-fluid variables

A two-fluid model assigns separate number densities n_i, n_e, velocities $\boldsymbol{v}_i, \boldsymbol{v}_e$, and pressures p_i, p_e to ions and electrons. A one-fluid model forms collective variables. For a singly ionized plasma with ion mass m_i much larger than electron mass m_e, representative definitions are

$$\rho_\mathrm{d} = m_i n_i + m_e n_e, \tag{3.1}$$

$$\rho_\mathrm{d} \boldsymbol{v}_\mathrm{d} = m_i n_i \boldsymbol{v}_i + m_e n_e \boldsymbol{v}_e, \tag{3.2}$$

$$\boldsymbol{j}_\mathrm{d} = e n_i \boldsymbol{v}_i - e n_e \boldsymbol{v}_e, \tag{3.3}$$

where $e > 0$ is the elementary charge, ρ_d is mass density, $\boldsymbol{v}_\mathrm{d}$ is center-of-mass fluid velocity, and $\boldsymbol{j}_\mathrm{d}$ is electric current density. The subscript d means *dimensional*; these quantities carry units from the International System of Units (SI). Later, unadorned project variables will be dimensionless.

The current density measures the relative motion of positive and negative charge. The bulk velocity measures the mass-weighted common motion. They are distinct even when the plasma is approximately quasi-neutral.

3.4 What is sacrificed in a fluid reduction?

Fluid MHD cannot, by itself, reproduce arbitrary velocity-space distributions, wave-particle resonances, finite-orbit effects, or collisionless kinetic damping. It is useful when the desired dynamics occurs on spatial and temporal scales sufficiently large compared with microscopic kinetic scales, and when a suitable closure represents the discarded moments. "MHD is valid" is never an absolute statement; it is a statement about the phenomena and scale range under study [1].

Learning checkpoint

The distribution function is a field on phase space. The MHD variables are also fields, but they live on ordinary space and time. MHD does not stop being a field theory when it stops explicitly evolving the distribution function.

4 Physical building blocks: electromagnetism and fluid conservation

4.1 Lorentz force

A particle of charge q and velocity $\boldsymbol{u}$ experiences the Lorentz force

$$\boldsymbol{F} = q\left(\boldsymbol{E}_\mathrm{d} + \boldsymbol{u} \times \boldsymbol{B}_\mathrm{d}\right), \tag{4.1}$$

where $\boldsymbol{E}_\mathrm{d}$ is the electric field and $\boldsymbol{B}_\mathrm{d}$ is the magnetic field. In a fluid, the corresponding macroscopic electromagnetic force density is

$$\boldsymbol{f}_\mathrm{EM} = \rho_{q,\mathrm{d}}\boldsymbol{E}_\mathrm{d} + \boldsymbol{j}_\mathrm{d} \times \boldsymbol{B}_\mathrm{d}, \tag{4.2}$$

where $\rho_{q,\mathrm{d}}$ is charge density. In nonrelativistic quasi-neutral MHD, the electric-force-density term is smaller in the standard ordering and the dominant macroscopic force is $\boldsymbol{j}_\mathrm{d} \times \boldsymbol{B}_\mathrm{d}$.

4.2 Reduced Maxwell equations used by MHD

The MHD model uses

$$\frac{\partial \boldsymbol{B}_\mathrm{d}}{\partial t_\mathrm{d}} = -\nabla_\mathrm{d} \times \boldsymbol{E}_\mathrm{d} \qquad \text{(Faraday's law)}, \tag{4.3}$$

$$\nabla_\mathrm{d} \times \boldsymbol{B}_\mathrm{d} = \mu_0 \boldsymbol{j}_\mathrm{d} \qquad \text{(Ampere's law without displacement current)}, \tag{4.4}$$

$$\nabla_\mathrm{d} \cdot \boldsymbol{B}_\mathrm{d} = 0 \qquad \text{(absence of magnetic monopoles)}. \tag{4.5}$$

Here μ_0 is the permeability of free space. The displacement current is neglected because standard MHD describes nonrelativistic motions with characteristic speeds much smaller than the speed of light [1].

4.3 Conservation of mass

For mass density ρ_d and fluid velocity $\boldsymbol{v}_\mathrm{d}$,

$$\frac{\partial \rho_\mathrm{d}}{\partial t_\mathrm{d}} + \nabla_\mathrm{d} \cdot (\rho_\mathrm{d} \boldsymbol{v}_\mathrm{d}) = 0. \tag{4.6}$$

Using the material derivative,

$$\frac{D\rho_\mathrm{d}}{Dt_\mathrm{d}} + \rho_\mathrm{d} \nabla_\mathrm{d} \cdot \boldsymbol{v}_\mathrm{d} = 0. \tag{4.7}$$

Thus density of a moving fluid element remains constant if the flow is incompressible.

4.4 Conservation of momentum

A simple viscous one-fluid momentum equation is

$$\rho_\mathrm{d} \left(\frac{\partial \boldsymbol{v}_\mathrm{d}}{\partial t_\mathrm{d}} + \boldsymbol{v}_\mathrm{d} \cdot \nabla_\mathrm{d} \boldsymbol{v}_\mathrm{d} \right) = -\nabla_\mathrm{d} p_\mathrm{d} + \boldsymbol{j}_\mathrm{d} \times \boldsymbol{B}_\mathrm{d} + \rho_\mathrm{d} \nu_\mathrm{d} \nabla_\mathrm{d}^2 \boldsymbol{v}_\mathrm{d}. \tag{4.8}$$

The scalar p_d is isotropic pressure and ν_d is kinematic viscosity, with units $\mathrm{m^2\,s^{-1}}$. The left-hand side is mass density times acceleration. On the right are the pressure-gradient force, Lorentz force, and a simplified viscous force.

Production extended-MHD models may use an anisotropic viscous stress tensor rather than the scalar Laplacian term in Eq. (4.8). The scalar form is a deliberate project assumption.

4.5 Thermodynamic closure

The fields ρ_d, $\boldsymbol{v}_\mathrm{d}$, p_d, and $\boldsymbol{B}_\mathrm{d}$ do not form a closed compressible system without an equation for pressure or internal energy. For example, an adiabatic ideal-gas closure has

$$\frac{Dp_\mathrm{d}}{Dt_\mathrm{d}} + \gamma p_\mathrm{d} \nabla_\mathrm{d} \cdot \boldsymbol{v}_\mathrm{d} = 0, \tag{4.9}$$

where γ is the ratio of specific heats. Resistive and viscous heating and thermal conduction add further terms.

The project removes this equation by imposing constant density and incompressibility and then taking the curl of the momentum equation. Pressure remains physically present as the field that enforces incompressibility, but it disappears from the vorticity evolution because the curl of a gradient is zero.

5 One-fluid resistive magnetohydrodynamics

5.1 Ohm's law in a moving conducting fluid

The simplest resistive Ohm's law is

$$\boldsymbol{E}_\mathrm{d} + \boldsymbol{v}_\mathrm{d} \times \boldsymbol{B}_\mathrm{d} = \rho_\Omega \boldsymbol{j}_\mathrm{d}. \tag{5.1}$$

The symbol ρ_Ω denotes *electrical resistivity*, with International System of Units (SI) units $\Omega\,\mathrm{m}$. It is the inverse of electrical conductivity. The combination $\boldsymbol{E}_\mathrm{d} + \boldsymbol{v}_\mathrm{d} \times \boldsymbol{B}_\mathrm{d}$ is the electric field measured in the moving-fluid frame in the nonrelativistic limit.

If $\rho_\Omega = 0$, Eq. (5.1) becomes the ideal-MHD condition

$$\boldsymbol{E}_\mathrm{d} + \boldsymbol{v}_\mathrm{d} \times \boldsymbol{B}_\mathrm{d} = 0. \tag{5.2}$$

Important distinction

Electrical resistivity ρ_Ω, dimensional magnetic diffusivity η_m, and dimensionless project diffusivity η are three related but different quantities:

$$\eta_\mathrm{m} = \frac{\rho_\Omega}{\mu_0}, \qquad \eta = \frac{\eta_\mathrm{m}}{L_0 V_A}.$$

The project symbol η never carries units of $\Omega\,\mathrm{m}$.

5.2 Derivation of the resistive induction equation

Insert Ohm's law into Faraday's law:

$$\frac{\partial \boldsymbol{B}_{\mathrm{d}}}{\partial t_{\mathrm{d}}} = -\nabla_{\mathrm{d}} \times \boldsymbol{E}_{\mathrm{d}} \tag{5.3}$$

$$= \nabla_{\mathrm{d}} \times (\boldsymbol{v}_{\mathrm{d}} \times \boldsymbol{B}_{\mathrm{d}}) - \nabla_{\mathrm{d}} \times (\rho_\Omega \boldsymbol{j}_{\mathrm{d}}). \tag{5.4}$$

For constant ρ_Ω, Ampere's law gives

$$-\nabla_{\mathrm{d}} \times (\rho_\Omega \boldsymbol{j}_{\mathrm{d}}) = -\frac{\rho_\Omega}{\mu_0}\nabla_{\mathrm{d}} \times (\nabla_{\mathrm{d}} \times \boldsymbol{B}_{\mathrm{d}}) \tag{5.5}$$

$$= \frac{\rho_\Omega}{\mu_0}\nabla_{\mathrm{d}}^2 \boldsymbol{B}_{\mathrm{d}}, \tag{5.6}$$

where $\nabla \cdot \boldsymbol{B} = 0$ was used. Defining magnetic diffusivity

$$\eta_{\mathrm{m}} \equiv \frac{\rho_\Omega}{\mu_0}, \tag{5.7}$$

we obtain

$$\frac{\partial \boldsymbol{B}_{\mathrm{d}}}{\partial t_{\mathrm{d}}} = \nabla_{\mathrm{d}} \times (\boldsymbol{v}_{\mathrm{d}} \times \boldsymbol{B}_{\mathrm{d}}) + \eta_{\mathrm{m}}\nabla_{\mathrm{d}}^2 \boldsymbol{B}_{\mathrm{d}}. \tag{5.8}$$

The first term transports and deforms magnetic field with the fluid. The second term diffuses magnetic field and permits magnetic topology to change. In an almost ideal plasma, η_{m} can be small while $\eta_{\mathrm{m}}\nabla^2 \boldsymbol{B}$ remains important inside a thin current sheet, where spatial derivatives become large [1, 2].

5.3 The one-fluid resistive-MHD set

In its simplified form, the parent model contains

$$\frac{\partial \rho_{\mathrm{d}}}{\partial t_{\mathrm{d}}} + \nabla_{\mathrm{d}} \cdot (\rho_{\mathrm{d}} \boldsymbol{v}_{\mathrm{d}}) = 0, \tag{5.9}$$

$$\rho_{\mathrm{d}}\frac{D\boldsymbol{v}_{\mathrm{d}}}{Dt_{\mathrm{d}}} = -\nabla_{\mathrm{d}} p_{\mathrm{d}} + \boldsymbol{j}_{\mathrm{d}} \times \boldsymbol{B}_{\mathrm{d}} + \rho_{\mathrm{d}}\nu_{\mathrm{d}}\nabla_{\mathrm{d}}^2 \boldsymbol{v}_{\mathrm{d}}, \tag{5.10}$$

$$\frac{\partial \boldsymbol{B}_{\mathrm{d}}}{\partial t_{\mathrm{d}}} = \nabla_{\mathrm{d}} \times (\boldsymbol{v}_{\mathrm{d}} \times \boldsymbol{B}_{\mathrm{d}}) + \eta_{\mathrm{m}}\nabla_{\mathrm{d}}^2 \boldsymbol{B}_{\mathrm{d}}, \tag{5.11}$$

$$\nabla_{\mathrm{d}} \cdot \boldsymbol{B}_{\mathrm{d}} = 0, \tag{5.12}$$

$$\mu_0 \boldsymbol{j}_{\mathrm{d}} = \nabla_{\mathrm{d}} \times \boldsymbol{B}_{\mathrm{d}}, \tag{5.13}$$

plus a thermodynamic closure. This is the physical parent from which the project model is reduced.

5.4 Ideal versus resistive MHD

Ideal MHD sets electrical resistivity and viscosity to zero. It is not the entirety of plasma physics. It is a particular nondissipative fluid limit. Resistive MHD retains finite electrical resistivity, producing Ohmic heating and magnetic diffusion. Viscous-resistive MHD also retains viscous diffusion of momentum or vorticity.

The word *non-ideal* is broader than *resistive*. Hall drift, electron-pressure terms, electron inertia, anisotropic stresses, and kinetic effects can all violate the ideal Ohm's law. The present project retains only scalar resistive and viscous departures from the ideal two-field dynamics.

6 Assumptions that define the project reduction

The final equations are not guessed. They follow from the parent model after the assumptions below are imposed. Each assumption has both a mathematical consequence and a physical cost.

Assumption	Mathematical consequence	Physics omitted or restricted
Two-dimensional Cartesian domain	$\partial_z = 0$; evolved fields depend on (x, y, t) only	Toroidal geometry, three-dimensional variation, and parallel structure
Constant mass density ρ_0	Density equation becomes redundant	Compression, density waves, stratification, and density transport
Incompressible flow	$\nabla_\perp \cdot \boldsymbol{v}_\perp = 0$	Fast and slow magnetosonic compression and acoustic dynamics
Strong, uniform guide field	$\boldsymbol{B} = B_g \hat{\boldsymbol{z}} + \boldsymbol{B}_\perp$	Evolution of guide-field magnitude and associated compressional physics
Perpendicular in-plane flow	$v_z = 0$	Parallel flow and parallel momentum dynamics
Scalar constant resistivity	One coefficient η_m multiplies $\nabla^2 \boldsymbol{B}$	Temperature-dependent, anisotropic, anomalous, and tensor resistivity
Scalar constant viscosity	One coefficient ν_d multiplies $\nabla^2 \boldsymbol{v}$	Anisotropic collisional stress, finite-gyroradius viscous corrections, and spatially varying transport
No separate pressure/energy evolution	Curl of momentum removes p	Temperature, heat conduction, Ohmic heating feedback, and pressure-driven modes

Assumption	Mathematical consequence	Physics omitted or restricted
Doubly periodic boundaries	No physical wall or boundary flux	Wall currents, vacuum region, separatrix, material interfaces, and sheath physics

6.1 Two-dimensional geometry and the guide field

Let the magnetic field be

$$\boldsymbol{B}_{\mathrm{d}} = B_g \widehat{\boldsymbol{z}} + \boldsymbol{B}_{\perp,\mathrm{d}}, \tag{6.1}$$

where B_g is a constant guide field and $\boldsymbol{B}_{\perp,\mathrm{d}}$ lies in the x-y plane. All fields satisfy $\partial_z = 0$. Because the guide field is spatially uniform, it produces no current. Because the current in this strict two-dimensional model points in the z direction, its cross product with $B_g\widehat{\boldsymbol{z}}$ vanishes. The guide field motivates the reduced ordering but does not explicitly appear in the final two-dimensional equations.

Important distinction

The full three-dimensional two-field reduced-MHD equations normally contain derivatives along the guide field. Setting $\partial_z = 0$ removes those terms. The project system is therefore the strictly two-dimensional limit of two-field reduced MHD and is also mathematically equivalent, up to sign and normalization conventions, to two-dimensional incompressible resistive MHD with a uniform guide field.

6.2 Constant density and incompressibility

Set $\rho_{\mathrm{d}} = \rho_0$, a positive constant. The continuity equation becomes

$$\rho_0 \nabla_{\mathrm{d}} \cdot \boldsymbol{v}_{\mathrm{d}} = 0, \tag{6.2}$$

so

$$\nabla_{\mathrm{d}} \cdot \boldsymbol{v}_{\mathrm{d}} = 0. \tag{6.3}$$

Incompressibility does not mean that the velocity is spatially uniform. It means that the local volume of a moving fluid element does not change. The flow may still shear, stretch, rotate, and form small scales.

6.3 Why pressure disappears but is not physically absent

Pressure in incompressible flow adjusts so that the velocity remains divergence-free. If one solves directly for velocity, pressure is a Lagrange-multiplier-like field coupled through an elliptic equation.

The project instead takes the curl of momentum. Since

$$\nabla \times (-\nabla p) = 0, \tag{6.4}$$

pressure disappears from the vorticity equation. This is a mathematical elimination of the pressure gradient, not an assertion that the plasma has zero pressure.

7 Flux and stream functions: building divergence-free fields

7.1 Magnetic-flux function

Define a dimensional scalar magnetic-flux function $\psi_{\mathrm{d}}(x_{\mathrm{d}}, y_{\mathrm{d}}, t_{\mathrm{d}})$ by

$$\boldsymbol{B}_{\perp,\mathrm{d}} = \nabla_{\mathrm{d}}\psi_{\mathrm{d}} \times \hat{\boldsymbol{z}}. \tag{7.1}$$

The cross product gives

$$\boldsymbol{B}_{\perp,\mathrm{d}} = \left(\frac{\partial \psi_{\mathrm{d}}}{\partial y_{\mathrm{d}}}, -\frac{\partial \psi_{\mathrm{d}}}{\partial x_{\mathrm{d}}}, 0\right). \tag{7.2}$$

Because $\boldsymbol{B}_{\perp,\mathrm{d}} = \nabla \times (\psi_{\mathrm{d}}\hat{\boldsymbol{z}})$, it is divergence-free automatically:

$$\nabla_{\mathrm{d}} \cdot \boldsymbol{B}_{\perp,\mathrm{d}} = 0. \tag{7.3}$$

The SI unit of ψ_{d} is tesla-meter, $\mathrm{T\,m}$, which is magnetic flux per unit distance in the ignorable z direction.

The magnetic field is tangent to contours of ψ_{d} because

$$\boldsymbol{B}_{\perp,\mathrm{d}} \cdot \nabla_{\mathrm{d}}\psi_{\mathrm{d}} = 0. \tag{7.4}$$

Thus, in a two-dimensional plot, contour lines of ψ are magnetic-field lines in the plane.

7.2 Flow stream function

Define a dimensional stream function U_{d} by

$$\boldsymbol{v}_{\perp,\mathrm{d}} = \hat{\boldsymbol{z}} \times \nabla_{\mathrm{d}} U_{\mathrm{d}}. \tag{7.5}$$

Therefore,

$$\boldsymbol{v}_{\perp,\mathrm{d}} = \left(-\frac{\partial U_{\mathrm{d}}}{\partial y_{\mathrm{d}}}, \frac{\partial U_{\mathrm{d}}}{\partial x_{\mathrm{d}}}, 0\right). \tag{7.6}$$

This field is also divergence-free identically. The SI unit of U_{d} is $\mathrm{m^2\,s^{-1}}$, because one spatial derivative of U_{d} must have velocity units.

Frozen project convention

After nondimensionalization, the project uses

$$\boxed{\boldsymbol{B}_\perp = \nabla\psi \times \hat{\boldsymbol{z}} = (\psi_y, -\psi_x), \qquad \boldsymbol{v}_\perp = \hat{\boldsymbol{z}} \times \nabla U = (-U_y, U_x).}$$

Changing either cross-product order changes signs throughout the model. These definitions are frozen.

7.3 Current density from the flux function

Ampere's law gives an out-of-plane current density. From Eq. (7.2),

$$\mu_0 j_{z,\mathrm{d}} = (\nabla_\mathrm{d} \times \boldsymbol{B}_{\perp,\mathrm{d}})_z \tag{7.7}$$

$$= \frac{\partial B_{y,\mathrm{d}}}{\partial x_\mathrm{d}} - \frac{\partial B_{x,\mathrm{d}}}{\partial y_\mathrm{d}} \tag{7.8}$$

$$= -\frac{\partial^2 \psi_\mathrm{d}}{\partial x_\mathrm{d}^2} - \frac{\partial^2 \psi_\mathrm{d}}{\partial y_\mathrm{d}^2} \tag{7.9}$$

$$= -\nabla^2_{\perp,\mathrm{d}} \psi_\mathrm{d}. \tag{7.10}$$

The dimensionless project current variable will be scaled so that

$$\boxed{J = -\nabla_\perp^2 \psi.} \tag{7.11}$$

The scalar J is proportional to physical current density $j_{z,\mathrm{d}}$; it is not itself an SI current density.

7.4 Vorticity from the stream function

The out-of-plane vorticity is

$$\omega_{z,\mathrm{d}} = (\nabla_\mathrm{d} \times \boldsymbol{v}_{\perp,\mathrm{d}})_z \tag{7.12}$$

$$= \frac{\partial v_{y,\mathrm{d}}}{\partial x_\mathrm{d}} - \frac{\partial v_{x,\mathrm{d}}}{\partial y_\mathrm{d}} \tag{7.13}$$

$$= \frac{\partial^2 U_\mathrm{d}}{\partial x_\mathrm{d}^2} + \frac{\partial^2 U_\mathrm{d}}{\partial y_\mathrm{d}^2} \tag{7.14}$$

$$= \nabla^2_{\perp,\mathrm{d}} U_\mathrm{d}. \tag{7.15}$$

After nondimensionalization,

$$\boxed{\omega = \nabla_\perp^2 U.} \tag{7.16}$$

Learning checkpoint

The signs are not arbitrary once the vector definitions are fixed. The chain is

$$\boldsymbol{B}_\perp = (\psi_y, -\psi_x) \Rightarrow J = -\nabla^2\psi, \qquad \boldsymbol{v}_\perp = (-U_y, U_x) \Rightarrow \omega = +\nabla^2 U.$$

The opposite signs of J and ω arise from the opposite cross-product order in the definitions of $\boldsymbol{B}_\perp$ and $\boldsymbol{v}_\perp$.

8 The Poisson bracket and two-dimensional advection

8.1 Definition

For two scalar fields $f(x, y)$ and $g(x, y)$, define the Poisson bracket

$$\boxed{[f, g] \equiv f_x g_y - f_y g_x.} \tag{8.1}$$

The name "Poisson bracket" is shared with Hamiltonian mechanics. Here it is a compact notation for a particular combination of two-dimensional derivatives.

8.2 The bracket is advection by the stream-function flow

Using $\boldsymbol{v}_\perp = (-U_y, U_x)$,

$$\begin{aligned} \boldsymbol{v}_\perp \cdot \nabla f &= (-U_y) f_x + (U_x) f_y && (8.2)\\ &= U_x f_y - U_y f_x && (8.3)\\ &= [U, f]\,. && (8.4) \end{aligned}$$

Therefore

$$\frac{Df}{Dt} = f_t + [U, f]\,. \tag{8.5}$$

The nonlinear terms $[U, \psi]$ and $[U, \omega]$ in the final equations are simply advection of magnetic flux and vorticity by the incompressible flow.

Similarly, using $\boldsymbol{B}_\perp = (\psi_y, -\psi_x)$,

$$\begin{aligned} \boldsymbol{B}_\perp \cdot \nabla f &= \psi_y f_x - \psi_x f_y && (8.6)\\ &= [f, \psi] = -\,[\psi, f]\,. && (8.7) \end{aligned}$$

8.3 Algebraic properties

The bracket satisfies:

$$[f, g] = -[g, f] \qquad \text{(antisymmetry)}, \tag{8.8}$$
$$[f, f] = 0, \tag{8.9}$$
$$[f, gh] = g\,[f, h] + h\,[f, g] \qquad \text{(product rule)}, \tag{8.10}$$
$$[f, [g, h]] + [g, [h, f]] + [h, [f, g]] = 0 \qquad \text{(Jacobi identity)}. \tag{8.11}$$

The Jacobi identity is structurally important in Hamiltonian formulations, although the first implementation does not need to evaluate it explicitly.

8.4 Integral identities on a periodic domain

Let $\Omega = [0, L_x) \times [0, L_y)$, with all fields periodic in both directions. Integration by parts produces no boundary contribution. Consequently,

$$\int_\Omega [f, g]\ \mathrm{d}A = 0, \tag{8.12}$$
$$\int_\Omega f\,[g, h]\ \mathrm{d}A = \int_\Omega g\,[h, f]\ \mathrm{d}A = \int_\Omega h\,[f, g]\ \mathrm{d}A. \tag{8.13}$$

Equation (8.13) is the cyclic bracket identity. It is central to the energy proof in Section 14.

To see why Eq. (8.12) holds, write

$$[f, g] = \frac{\partial}{\partial x}(f g_y) - \frac{\partial}{\partial y}(f g_x). \tag{8.14}$$

The integral of each total derivative across one periodic direction vanishes because the values at the two identified faces are equal.

9 Derivation of the magnetic-flux equation

The first project equation is

$$\psi_t + [U, \psi] = \eta \nabla_\perp^2 \psi. \tag{9.1}$$

We derive it first in dimensional form, then nondimensionalize it in Section 11.

9.1 Vector-potential gauge used in two dimensions

Because

$$\boldsymbol{B}_{\perp,\mathrm{d}} = \nabla_\mathrm{d} \times (\psi_\mathrm{d} \hat{\boldsymbol{z}}), \tag{9.2}$$

we may use a vector potential $\boldsymbol{A}_\mathrm{d} = \psi_\mathrm{d}\hat{\boldsymbol{z}}$. The electric field can be written

$$\boldsymbol{E}_\mathrm{d} = -\frac{\partial \boldsymbol{A}_\mathrm{d}}{\partial t_\mathrm{d}} - \nabla_\mathrm{d}\Phi_\mathrm{d}, \tag{9.3}$$

where Φ_d is electrostatic potential. Since $\partial_z \Phi_\mathrm{d} = 0$, the out-of-plane component is

$$E_{z,\mathrm{d}} = -\frac{\partial \psi_\mathrm{d}}{\partial t_\mathrm{d}}. \tag{9.4}$$

9.2 The out-of-plane component of Ohm's law

The z component of $\boldsymbol{v} \times \boldsymbol{B}$ is

$$(\boldsymbol{v}_{\perp,\mathrm{d}} \times \boldsymbol{B}_{\perp,\mathrm{d}})_z = v_{x,\mathrm{d}} B_{y,\mathrm{d}} - v_{y,\mathrm{d}} B_{x,\mathrm{d}} \tag{9.5}$$

$$= (-U_{\mathrm{d},y})(-\psi_{\mathrm{d},x}) - (U_{\mathrm{d},x})(\psi_{\mathrm{d},y}) \tag{9.6}$$

$$= -\left[U_\mathrm{d}, \psi_\mathrm{d}\right]_\mathrm{d}. \tag{9.7}$$

The subscript on the bracket indicates derivatives with respect to dimensional coordinates. The z component of resistive Ohm's law is therefore

$$-\psi_{\mathrm{d},t} - \left[U_\mathrm{d}, \psi_\mathrm{d}\right]_\mathrm{d} = \rho_\Omega j_{z,\mathrm{d}}. \tag{9.8}$$

Using $\mu_0 j_{z,\mathrm{d}} = -\nabla^2_{\perp,\mathrm{d}}\psi_\mathrm{d}$ and $\eta_\mathrm{m} = \rho_\Omega/\mu_0$ gives

$$\boxed{\frac{\partial \psi_\mathrm{d}}{\partial t_\mathrm{d}} + \left[U_\mathrm{d}, \psi_\mathrm{d}\right]_\mathrm{d} = \eta_\mathrm{m}\nabla^2_{\perp,\mathrm{d}}\psi_\mathrm{d}.} \tag{9.9}$$

9.3 Physical interpretation

The left-hand side is the material derivative of ψ_d:

$$\frac{D\psi_\mathrm{d}}{Dt_\mathrm{d}} = \frac{\partial \psi_\mathrm{d}}{\partial t_\mathrm{d}} + \boldsymbol{v}_{\perp,\mathrm{d}} \cdot \nabla_\mathrm{d}\psi_\mathrm{d}. \tag{9.10}$$

Thus

$$\frac{D\psi_\mathrm{d}}{Dt_\mathrm{d}} = \eta_\mathrm{m}\nabla^2_{\perp,\mathrm{d}}\psi_\mathrm{d}. \tag{9.11}$$

In the ideal limit $\eta_\mathrm{m} = 0$, each moving fluid element retains its value of ψ_d. This is the two-dimensional expression of frozen-in magnetic flux. At finite resistivity, diffusion changes the flux carried by a fluid element.

Using $J = -\nabla^2\psi$, the dimensionless resistive source is

$$\eta\nabla^2\psi = -\eta J. \tag{9.12}$$

Therefore resistive evolution is strongest where the magnitude of the current variable is large.

10 Derivation of the vorticity equation

The second project equation is

$$\omega_t + [U, \omega] = [J, \psi] + \nu \nabla_\perp^2 \omega. \tag{10.1}$$

It is the out-of-plane curl of the momentum equation.

10.1 Begin with incompressible momentum balance

With constant density ρ_0,

$$\frac{\partial \boldsymbol{v}_\mathrm{d}}{\partial t_\mathrm{d}} + \boldsymbol{v}_\mathrm{d} \cdot \nabla_\mathrm{d} \boldsymbol{v}_\mathrm{d} = -\frac{1}{\rho_0} \nabla_\mathrm{d} p_\mathrm{d} + \frac{1}{\rho_0} \boldsymbol{j}_\mathrm{d} \times \boldsymbol{B}_\mathrm{d} + \nu_\mathrm{d} \nabla_\mathrm{d}^2 \boldsymbol{v}_\mathrm{d}. \tag{10.2}$$

Take $\nabla_\mathrm{d}\times$ of every term and select the z component.

10.2 Time derivative

Curl and the time derivative commute for smooth fields:

$$\left[\nabla_\mathrm{d} \times \frac{\partial \boldsymbol{v}_\mathrm{d}}{\partial t_\mathrm{d}}\right]_z = \frac{\partial \omega_{z,\mathrm{d}}}{\partial t_\mathrm{d}}. \tag{10.3}$$

10.3 Nonlinear inertia becomes vorticity advection

For incompressible two-dimensional flow,

$$[\nabla_\mathrm{d} \times (\boldsymbol{v}_\mathrm{d} \cdot \nabla_\mathrm{d} \boldsymbol{v}_\mathrm{d})]_z = \boldsymbol{v}_\mathrm{d} \cdot \nabla_\mathrm{d} \omega_{z,\mathrm{d}}. \tag{10.4}$$

Using the stream function,

$$\boldsymbol{v}_\mathrm{d} \cdot \nabla_\mathrm{d} \omega_{z,\mathrm{d}} = [U_\mathrm{d}, \omega_{z,\mathrm{d}}]_\mathrm{d}\,. \tag{10.5}$$

One route to the identity is the vector decomposition

$$(\boldsymbol{v} \cdot \nabla)\boldsymbol{v} = \nabla \left(\frac{|\boldsymbol{v}|^2}{2}\right) - \boldsymbol{v} \times \boldsymbol{\omega}. \tag{10.6}$$

The first term has zero curl; the second produces vorticity transport in the two-dimensional incompressible limit.

10.4 Pressure vanishes under curl

Because ρ_0 is constant,

$$\nabla_\mathrm{d} \times \left(-\frac{1}{\rho_0}\nabla_\mathrm{d} p_\mathrm{d}\right) = 0. \tag{10.7}$$

If density were spatially variable, the curl of $-(1/\rho)\nabla p$ would generally not vanish. It would contain *baroclinic vorticity generation*, which occurs when pressure and density gradients are not parallel. Constant density is therefore essential to this simple reduction.

10.5 Lorentz force becomes the magnetic bracket

The current is $\boldsymbol{j}_\mathrm{d} = j_{z,\mathrm{d}}\widehat{\boldsymbol{z}}$ and the in-plane magnetic field is $\boldsymbol{B}_{\perp,\mathrm{d}} = (\psi_y, -\psi_x, 0)$. Hence

$$\begin{aligned}\boldsymbol{j}_\mathrm{d} \times \boldsymbol{B}_{\perp,\mathrm{d}} &= j_{z,\mathrm{d}}\widehat{\boldsymbol{z}} \times (\psi_y\widehat{\boldsymbol{x}} - \psi_x\widehat{\boldsymbol{y}}) && (10.8)\\ &= j_{z,\mathrm{d}}\nabla_\mathrm{d}\psi_\mathrm{d}. && (10.9)\end{aligned}$$

Its out-of-plane curl is

$$\begin{aligned}\left[\nabla_\mathrm{d} \times (j_{z,\mathrm{d}}\nabla_\mathrm{d}\psi_\mathrm{d})\right]_z &= \frac{\partial j_{z,\mathrm{d}}}{\partial x_\mathrm{d}}\frac{\partial \psi_\mathrm{d}}{\partial y_\mathrm{d}} - \frac{\partial j_{z,\mathrm{d}}}{\partial y_\mathrm{d}}\frac{\partial \psi_\mathrm{d}}{\partial x_\mathrm{d}} && (10.10)\\ &= \left[j_{z,\mathrm{d}}, \psi_\mathrm{d}\right]_\mathrm{d}. && (10.11)\end{aligned}$$

The Lorentz force therefore generates vorticity through spatial misalignment of current-density and flux contours. If $j_{z,\mathrm{d}}$ is a function of ψ_d alone, then their gradients are parallel and the bracket vanishes.

10.6 Viscosity diffuses vorticity

For constant ν_d,

$$\left[\nabla_\mathrm{d} \times (\nu_\mathrm{d}\nabla^2_\mathrm{d}\boldsymbol{v}_\mathrm{d})\right]_z = \nu_\mathrm{d}\nabla^2_{\perp,\mathrm{d}}\omega_{z,\mathrm{d}}. \tag{10.12}$$

10.7 Dimensional vorticity equation

Combining the terms gives

$$\boxed{\frac{\partial \omega_{z,\mathrm{d}}}{\partial t_\mathrm{d}} + \left[U_\mathrm{d}, \omega_{z,\mathrm{d}}\right]_\mathrm{d} = \frac{1}{\rho_0}\left[j_{z,\mathrm{d}}, \psi_\mathrm{d}\right]_\mathrm{d} + \nu_\mathrm{d}\nabla^2_{\perp,\mathrm{d}}\omega_{z,\mathrm{d}}.} \tag{10.13}$$

Alfvén normalization will turn the coefficient of the magnetic bracket into unity.

Learning checkpoint

The two nonlinear brackets have different jobs:

- $[U, \omega]$ transports existing vorticity with the fluid;
- $[J, \psi]$ is the curl of the Lorentz force and can generate or redistribute vorticity.

Neither bracket is a dissipative term. Their domain integrals vanish under periodic boundary conditions.

11 Nondimensionalization and characteristic time scales

11.1 Why nondimensionalize?

Nondimensionalization replaces quantities carrying SI units with ratios to chosen reference scales. It has four benefits here:

(a) it exposes the relative strength of physical mechanisms;

(b) it reduces the number of independent coefficients;

(c) it makes parameter ranges portable between physical systems with similar dimensionless numbers; and

(d) it places neural-network inputs and outputs on more manageable physical scales.

Nondimensionalization is a physical rescaling. It is not the same as later machine-learning standardization or min-max scaling.

11.2 Reference scales

Choose positive dimensional scales:

- L_0: characteristic perpendicular length;
- B_0: characteristic in-plane magnetic-field magnitude;
- ρ_0: constant mass density;
- V_A: Alfvén speed; and
- t_A: Alfvén time.

The Alfvén speed is

$$V_A \equiv \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \tag{11.1}$$

and the Alfvén time is

$$t_A \equiv \frac{L_0}{V_A}. \tag{11.2}$$

It is the time for a disturbance traveling at V_A to traverse the reference length L_0.

11.3 Dimensionless variables

Define

$$x_\mathrm{d} = L_0 x, \qquad y_\mathrm{d} = L_0 y, \qquad t_\mathrm{d} = t_A t, \tag{11.3}$$

$$\psi_\mathrm{d} = B_0 L_0 \psi, \qquad U_\mathrm{d} = V_A L_0 U, \tag{11.4}$$

$$\boldsymbol{B}_{\perp,\mathrm{d}} = B_0 \boldsymbol{B}_\perp, \qquad \boldsymbol{v}_{\perp,\mathrm{d}} = V_A \boldsymbol{v}_\perp, \tag{11.5}$$

$$j_{z,\mathrm{d}} = \frac{B_0}{\mu_0 L_0} J, \qquad \omega_{z,\mathrm{d}} = \frac{V_A}{L_0}\omega. \tag{11.6}$$

Spatial and temporal derivatives transform as

$$\nabla_\mathrm{d} = \frac{1}{L_0}\nabla, \qquad \frac{\partial}{\partial t_\mathrm{d}} = \frac{1}{t_A}\frac{\partial}{\partial t}. \tag{11.7}$$

Substitution into Eq. (7.10) gives

$$J = -\nabla_\perp^2 \psi, \tag{11.8}$$

and substitution into the vorticity definition gives

$$\omega = \nabla_\perp^2 U. \tag{11.9}$$

11.4 Dimensionless diffusivities

The flux equation becomes

$$\psi_t + [U, \psi] = \underbrace{\frac{\eta_\mathrm{m}}{L_0 V_A}}_{\eta} \nabla_\perp^2 \psi. \tag{11.10}$$

Thus

$$\boxed{\eta \equiv \frac{\eta_\mathrm{m}}{L_0 V_A}.} \tag{11.11}$$

For the magnetic term in the vorticity equation, use $B_0^2/(\mu_0\rho_0) = V_A^2$. The coefficient becomes exactly one. The viscous coefficient is

$$\boxed{\nu \equiv \frac{\nu_\mathrm{d}}{L_0 V_A}.} \tag{11.12}$$

11.5 Lundquist, Reynolds, and magnetic Prandtl numbers

The Alfvénic Lundquist number is

$$S \equiv \frac{L_0 V_A}{\eta_{\mathrm{m}}} = \frac{1}{\eta}. \tag{11.13}$$

The fluid Reynolds number based on V_A is

$$\mathrm{Re} \equiv \frac{L_0 V_A}{\nu_{\mathrm{d}}} = \frac{1}{\nu}. \tag{11.14}$$

The magnetic Prandtl number is

$$\mathrm{Pm} \equiv \frac{\nu_{\mathrm{d}}}{\eta_{\mathrm{m}}} = \frac{\nu}{\eta} = \frac{S}{\mathrm{Re}}. \tag{11.15}$$

A magnetic Reynolds number based on an actual characteristic flow speed V_0 is

$$\mathrm{Rm} \equiv \frac{L_0 V_0}{\eta_{\mathrm{m}}}. \tag{11.16}$$

It equals S only when $V_0 = V_A$. The project uses Alfvén normalization, so $1/\eta$ is most precisely called the Lundquist number, although literature sometimes loosely refers to it as a magnetic Reynolds number.

11.6 Time-scale interpretation

The dimensional diffusion times are

$$t_\eta = \frac{L_0^2}{\eta_{\mathrm{m}}}, \tag{11.17}$$

$$t_\nu = \frac{L_0^2}{\nu_{\mathrm{d}}}. \tag{11.18}$$

Relative to t_A,

$$\frac{t_\eta}{t_A} = S = \frac{1}{\eta}, \qquad \frac{t_\nu}{t_A} = \mathrm{Re} = \frac{1}{\nu}. \tag{11.19}$$

Scale	Expression	Meaning
Alfvén time	$t_A = L_0/V_A$	Ideal magnetic-tension response across L_0
Resistive time	$t_\eta = L_0^2/\eta_{\mathrm{m}}$	Diffusive smoothing of magnetic structure on scale L_0
Viscous time	$t_\nu = L_0^2/\nu_{\mathrm{d}}$	Diffusive smoothing of velocity on scale L_0
Lundquist number	$S = t_\eta/t_A$	Separation between resistive and Alfvénic times

Scale	Expression	Meaning
Reynolds number	$\mathrm{Re} = t_\nu / t_A$	Separation between viscous and Alfvénic times

If $\eta = 10^{-3}$ and $\nu = 10^{-3}$, then $S = \mathrm{Re} = 10^3$ and $\mathrm{Pm} = 1$ for the chosen reference scales. This does not mean that every structure waits $10^3 t_A$ to diffuse. A current sheet of dimensionless thickness $\delta \ll 1$ has a local resistive time of order δ^2/η, which can be much shorter.

11.7 The dimensionless domain used by version 1

The frozen project domain is

$$\Omega = [0, L_x) \times [0, L_y), \qquad L_x = L_y = 2\pi \tag{11.20}$$

for the baseline case. Because $x = x_\mathrm{d}/L_0$, a dimensionless box of length 2π corresponds to physical length $2\pi L_0$. The choice is convenient for Fourier modes with integer wavenumbers. It is not a statement that a fusion device has square Cartesian geometry.

11.8 Physical nondimensionalization versus neural input scaling

Suppose a neural network uses

$$\xi = \frac{2x}{L_x} - 1, \qquad \tau = \frac{2t}{T} - 1 \tag{11.21}$$

to map a dimensionless physical interval to $[-1, 1]$. Then automatic derivatives with respect to network coordinates must be converted by the chain rule:

$$\frac{\partial}{\partial x} = \frac{2}{L_x}\frac{\partial}{\partial \xi}, \qquad \frac{\partial}{\partial t} = \frac{2}{T}\frac{\partial}{\partial \tau}. \tag{11.22}$$

Forgetting these factors changes the PDE. The physical variables are already nondimensional before this additional numerical transformation.

12 The frozen two-field model

12.1 Domain and fields

The independent variables are

$$(x, y, t) \in [0, L_x) \times [0, L_y) \times [0, T]. \tag{12.1}$$

Here $T > 0$ is the final dimensionless time of the modeled interval. The dataset and neural state use ψ and U as primary fields. Current and vorticity are derived:

$$J = -\nabla_\perp^2 \psi, \qquad \omega = \nabla_\perp^2 U. \tag{12.2}$$

12.2 Evolution equations

The version-1 governing equations are

$$\boxed{\psi_t + [U, \psi] = \eta \nabla_\perp^2 \psi,} \tag{12.3}$$

$$\boxed{\omega_t + [U, \omega] = [J, \psi] + \nu \nabla_\perp^2 \omega.} \tag{12.4}$$

Together with Eq. (12.2), these form a closed two-field system.

Frozen project convention

The complete version-1 sign convention is

$$\begin{aligned}
[f, g] &= f_x g_y - f_y g_x, \\
\boldsymbol{B}_\perp &= \nabla\psi \times \hat{\boldsymbol{z}} = (\psi_y, -\psi_x), \\
\boldsymbol{v}_\perp &= \hat{\boldsymbol{z}} \times \nabla U = (-U_y, U_x), \\
J &= -\nabla_\perp^2 \psi, \\
\omega &= +\nabla_\perp^2 U, \\
\psi_t + [U, \psi] &= \eta \nabla_\perp^2 \psi, \\
\omega_t + [U, \omega] &= [J, \psi] + \nu \nabla_\perp^2 \omega.
\end{aligned}$$

Any stored dataset or model using different signs requires an explicit schema or convention change.

12.3 Term-by-term interpretation

Term	Mathematical role	Physical meaning
ψ_t	Local time change	Change of magnetic flux at a fixed point
$[U, \psi]$	Nonlinear advection	Transport of magnetic-flux contours by the fluid
$\eta \nabla^2 \psi = -\eta J$	Magnetic diffusion	Resistive slippage of field relative to fluid; topology change in current layers
ω_t	Local time change	Change of vorticity at a fixed point

Term	Mathematical role	Physical meaning
$[U, \omega]$	Nonlinear advection	Transport of vorticity by the incompressible flow
$[J, \psi]$	Magnetic forcing	Curl of the Lorentz force; converts magnetic and kinetic energy
$\nu\nabla^2\omega$	Viscous diffusion	Smoothing of vorticity and dissipation of kinetic energy

12.4 Primary fields versus prognostic variables

The physical state contract stores and predicts (ψ, U), from which (J, ω) are derived. A conventional pseudo-spectral solver may nevertheless time-advance (ψ, ω) because Eq. (12.4) directly gives ω_t. It then solves the periodic Poisson equation

$$\nabla_\perp^2 U = \omega \tag{12.5}$$

at each time stage. This numerical choice does not alter the field definitions. It only distinguishes the *dataset primary fields* from the solver's most convenient *prognostic variables*.

12.5 Useful limiting cases

Limit	Coefficients or fields	Interpretation
Ideal two-field MHD	$\eta = \nu = 0$	No explicit resistive or viscous dissipation; flux is materially conserved
Resistive, inviscid	$\eta > 0, \nu = 0$	Magnetic energy dissipates; kinetic small scales are not directly damped
Ideal magnetic, viscous	$\eta = 0, \nu > 0$	Magnetic topology remains frozen while flow dissipates
Pure magnetic diffusion	$U = 0$ and $[J, \psi] = 0$	$\psi_t = \eta\nabla^2\psi$
Pure viscous decay	$\psi = 0$	$\omega_t + [U, \omega] = \nu\nabla^2\omega$; a single Fourier mode has no self-bracket
Static ideal equilibrium	$U = 0,\ \eta = \nu = 0,\ [J, \psi] = 0$	Lorentz force has zero curl; J may be any single-valued function of ψ

13 Periodic boundaries, gauge freedom, and zero modes

13.1 Doubly periodic boundary conditions

The domain identifies opposite faces. For any evolved field f,

$$f(0, y, t) = f(L_x, y, t), \tag{13.1}$$
$$f(x, 0, t) = f(x, L_y, t). \tag{13.2}$$

The derivatives required by the PDE must also match periodically. A Fourier representation enforces smooth periodicity by construction.

Periodicity means there is no physical wall. Energy cannot leave through a boundary in the mathematical model. Any decrease in total energy must therefore come from the explicit resistive and viscous terms, up to numerical error.

13.2 Gauge freedom of the stream function

Velocity depends only on derivatives of U. Therefore

$$U(x, y, t) \mapsto U(x, y, t) + C_U(t) \tag{13.3}$$

leaves $\boldsymbol{v}$ and ω unchanged for any spatially uniform $C_U(t)$. The project fixes this nonuniqueness by imposing

$$\boxed{\langle U \rangle (t) = 0,} \tag{13.4}$$

where the spatial mean is

$$\langle f \rangle \equiv \frac{1}{|\Omega|} \int_\Omega f \, \mathrm{d}A, \qquad |\Omega| = L_x L_y. \tag{13.5}$$

Here $\mathrm{d}A = \mathrm{d}x\, \mathrm{d}y$ is the differential area element.

In Fourier space, a hat denotes a Fourier coefficient and $\boldsymbol{k} = (k_x, k_y)$ denotes its wavevector. The equation $\nabla^2 U = \omega$ becomes

$$-|\boldsymbol{k}|^2 \widehat{U}_{\boldsymbol{k}} = \widehat{\omega}_{\boldsymbol{k}}. \tag{13.6}$$

For every nonzero wavevector $\boldsymbol{k}$,

$$\widehat{U}_{\boldsymbol{k}} = -\frac{\widehat{\omega}_{\boldsymbol{k}}}{|\boldsymbol{k}|^2}. \tag{13.7}$$

At $\boldsymbol{k} = 0$, division by $|\boldsymbol{k}|^2$ is impossible. Equation (13.4) sets $\widehat{U}_{\boldsymbol{0}} = 0$. A periodic Laplacian has zero mean, so compatible vorticity satisfies $\widehat{\omega}_{\boldsymbol{0}} = \langle \omega \rangle = 0$.

13.3 Gauge and mean value of the flux function

The in-plane magnetic field is also unchanged by adding a spatial constant to ψ. In the chosen electric-potential gauge, the evolution equation preserves the spatial mean:

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\psi\rangle = -\langle[U,\psi]\rangle + \eta\left\langle\nabla^2\psi\right\rangle \tag{13.8}$$

$$= 0. \tag{13.9}$$

The project records the initial mean of ψ and maintains it across a case. Setting it to zero is convenient but not logically required if metadata retains the chosen value.

13.4 Consequences of strict periodicity

For a smooth periodic ψ,

$$\langle J\rangle = -\left\langle\nabla^2\psi\right\rangle = 0, \tag{13.10}$$

and the mean in-plane magnetic field is also zero because it is made from derivatives of a periodic scalar. A nonzero uniform guide field $B_g\hat{\boldsymbol{z}}$ is represented separately and is not part of ψ.

If a future problem requires a nonzero uniform in-plane field, one may decompose the flux function into a nonperiodic linear background plus a periodic perturbation. That is not part of version 1 and would require an explicit convention update.

13.5 Initial-condition compatibility

At $t = 0$, one must specify periodic fields

$$\psi(x,y,0) = \psi_0(x,y), \qquad U(x,y,0) = U_0(x,y), \tag{13.11}$$

or equivalently ψ_0 and ω_0 together with the zero-mean gauge for U_0. The initial current and vorticity are not independent if the dataset contract declares them derived:

$$J_0 = -\nabla^2\psi_0, \qquad \omega_0 = \nabla^2 U_0. \tag{13.12}$$

Supplying four arbitrary arrays would generally violate the model.

14 Energy, invariants, and dissipation

Conservation laws are not decorative theory. They are some of the strongest tests of a nonlinear solver and some of the most meaningful global constraints for a PINN.

14.1 Magnetic and kinetic energy

In dimensionless Alfvén units, define

$$E_B(t) \equiv \frac{1}{2}\int_\Omega |\boldsymbol{B}_\perp|^2 \, \mathrm{d}A = \frac{1}{2}\int_\Omega |\nabla\psi|^2 \, \mathrm{d}A, \tag{14.1}$$

$$E_K(t) \equiv \frac{1}{2}\int_\Omega |\boldsymbol{v}_\perp|^2 \, \mathrm{d}A = \frac{1}{2}\int_\Omega |\nabla U|^2 \, \mathrm{d}A. \tag{14.2}$$

The guide-field energy is constant and omitted from the dynamic energy budget.

14.2 Magnetic-energy equation

Differentiate Eq. (14.1) and integrate by parts:

$$\frac{\mathrm{d}E_B}{\mathrm{d}t} = \int_\Omega \nabla\psi \cdot \nabla\psi_t \, \mathrm{d}A \tag{14.3}$$

$$= -\int_\Omega (\nabla^2\psi)\psi_t \, \mathrm{d}A \tag{14.4}$$

$$= \int_\Omega J\psi_t \, \mathrm{d}A. \tag{14.5}$$

The flux equation gives $\psi_t = -[U, \psi] - \eta J$, so

$$\frac{\mathrm{d}E_B}{\mathrm{d}t} = -\int_\Omega J[U,\psi] \, \mathrm{d}A - \eta\int_\Omega J^2 \, \mathrm{d}A \tag{14.6}$$

$$= \int_\Omega U[J,\psi] \, \mathrm{d}A - \eta\int_\Omega J^2 \, \mathrm{d}A, \tag{14.7}$$

where the cyclic bracket identity was used.

14.3 Kinetic-energy equation

Integration by parts gives

$$E_K = -\frac{1}{2}\int_\Omega U\omega \, \mathrm{d}A. \tag{14.8}$$

Because $\omega = \nabla^2 U$ and the Laplacian is self-adjoint on a periodic domain,

$$\frac{\mathrm{d}E_K}{\mathrm{d}t} = -\int_\Omega U\omega_t \, \mathrm{d}A. \tag{14.9}$$

Insert the vorticity equation:

$$\frac{\mathrm{d}E_K}{\mathrm{d}t} = \int_\Omega U[U,\omega] \, \mathrm{d}A - \int_\Omega U[J,\psi] \, \mathrm{d}A - \nu\int_\Omega U\nabla^2\omega \, \mathrm{d}A \tag{14.10}$$

$$= -\int_\Omega U[J,\psi] \, \mathrm{d}A - \nu\int_\Omega \omega^2 \, \mathrm{d}A. \tag{14.11}$$

The self-advection integral vanishes, and

$$\int_\Omega U\nabla^2\omega\,\mathrm{d}A = \int_\Omega (\nabla^2 U)\omega\,\mathrm{d}A = \int_\Omega \omega^2\,\mathrm{d}A. \tag{14.12}$$

14.4 Total-energy law

Adding Eqs. (14.7) and (14.11), the magnetic-to-kinetic exchange term cancels:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}(E_B + E_K) = -\eta\int_\Omega J^2\,\mathrm{d}A - \nu\int_\Omega \omega^2\,\mathrm{d}A.} \tag{14.13}$$

For $\eta \geq 0$ and $\nu \geq 0$, total energy cannot increase in the unforced periodic model. In the ideal limit, total energy is conserved. The exchange term

$$\mathcal{T}_{B\leftrightarrow K} \equiv \int_\Omega U[J,\psi]\,\mathrm{d}A \tag{14.14}$$

appears with opposite signs in the two component budgets. It transfers energy between magnetic and kinetic reservoirs without creating or destroying total energy.

Frozen project convention

Equation (14.13) is a sign audit. If a code using the frozen definitions produces

$$\frac{\mathrm{d}E}{\mathrm{d}t} = +\eta\int J^2\,\mathrm{d}A \quad \text{or} \quad +\nu\int \omega^2\,\mathrm{d}A,$$

then at least one current, vorticity, diffusion, or evolution-equation sign is inconsistent.

14.5 Mean-square magnetic potential

Define

$$\mathcal{A}(t) = \frac{1}{2}\int_\Omega \psi^2\,\mathrm{d}A. \tag{14.15}$$

Then

$$\frac{\mathrm{d}\mathcal{A}}{\mathrm{d}t} = \int_\Omega \psi\left(-[U,\psi] + \eta\nabla^2\psi\right)\,\mathrm{d}A \tag{14.16}$$

$$= -\eta\int_\Omega |\nabla\psi|^2\,\mathrm{d}A \tag{14.17}$$

$$= -2\eta E_B. \tag{14.18}$$

Thus $\int \psi^2 \mathrm{d}A$ is an ideal invariant and decays resistively when $\eta > 0$.

More generally, when $\eta = 0$, every integral $\int_\Omega F(\psi)\mathrm{d}A$ is conserved for a sufficiently smooth function F, because incompressible advection rearranges values of ψ without changing their area distribution.

14.6 Cross helicity in two dimensions

Define cross helicity

$$H_C \equiv \int_\Omega \boldsymbol{v}_\perp \cdot \boldsymbol{B}_\perp \,\mathrm{d}A \tag{14.19}$$

$$= -\int_\Omega \nabla U \cdot \nabla\psi \,\mathrm{d}A \tag{14.20}$$

$$= \int_\Omega \psi\omega \,\mathrm{d}A = -\int_\Omega UJ \,\mathrm{d}A. \tag{14.21}$$

Direct differentiation gives

$$\frac{\mathrm{d}H_C}{\mathrm{d}t} = -(\eta+\nu)\int_\Omega J\omega \,\mathrm{d}A. \tag{14.22}$$

Therefore H_C is conserved in the ideal model. Unlike total energy, it need not decay monotonically for finite η and ν because $J\omega$ need not be positive everywhere.

14.7 Vorticity and current means

Because both ω and J are periodic Laplacians,

$$\int_\Omega \omega \,\mathrm{d}A = 0, \qquad \int_\Omega J \,\mathrm{d}A = 0. \tag{14.23}$$

These identities are compatibility conditions, not merely approximate conservation laws.

14.8 Enstrophy

The kinetic enstrophy is

$$Z_\omega = \frac{1}{2}\int_\Omega \omega^2 \,\mathrm{d}A. \tag{14.24}$$

Its budget is

$$\frac{\mathrm{d}Z_\omega}{\mathrm{d}t} = \int_\Omega \omega[J,\psi]\,\mathrm{d}A - \nu\int_\Omega |\nabla\omega|^2 \,\mathrm{d}A. \tag{14.25}$$

The Lorentz term can generate vorticity gradients, while viscosity dissipates them. Unlike unmagnetized two-dimensional Euler flow, vorticity enstrophy is not generally an ideal invariant here because magnetic forcing is present.

15 Magnetic topology, current sheets, and reconnection

15.1 Flux contours are field lines

Since $\boldsymbol{B}_\perp \cdot \nabla\psi = 0$, the magnetic field is tangent to every constant-ψ contour. Magnetic topology in two dimensions is therefore the topology of the contour map of ψ.

A point where $\nabla\psi = 0$ is a magnetic critical point. Near such a point (x_c, y_c), Taylor expansion gives

$$\psi \approx \psi_c + \frac{1}{2}\begin{bmatrix}\delta x & \delta y\end{bmatrix} \boldsymbol{H}_\psi \begin{bmatrix}\delta x \\ \delta y\end{bmatrix}, \tag{15.1}$$

where $\delta x = x - x_c$, $\delta y = y - y_c$, and $\boldsymbol{H}_\psi$ is the Hessian matrix of second derivatives.

- If the two Hessian eigenvalues have the same sign, the point is a local extremum. Nearby contours are closed, and the point is an **O-point**.
- If the eigenvalues have opposite signs, the point is a saddle. The crossing contour is a **separatrix**, and the point is an **X-point**.

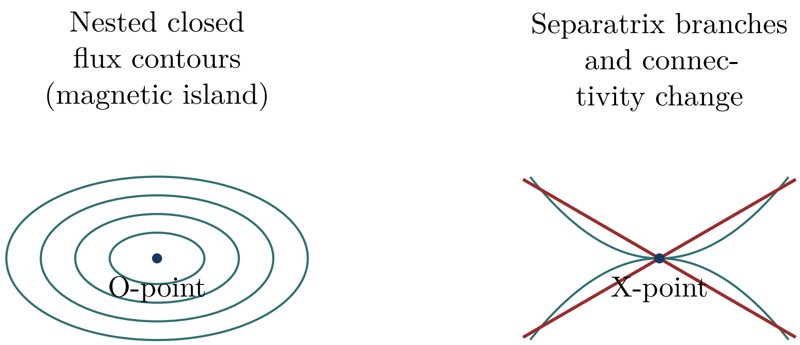

Figure 2: Conceptual contour structures. Actual project contours need not be ellipses or parabolas. An island is a region of closed magnetic-flux contours around an O-point; an X-point lies on the separatrix.

15.2 Frozen topology in the ideal model

When $\eta = 0$,

$$\frac{D\psi}{Dt} = 0. \tag{15.2}$$

Every fluid element preserves its value of ψ. Because the flow is smooth and incompressible, it continuously deforms flux contours but cannot cut and reconnect them. Magnetic field lines are said to be *frozen into* the fluid.

Flux freezing is a statement about topology and material transport, not a statement that the magnetic field is spatially or temporally constant. Ideal motion can stretch field lines, rotate them, amplify gradients, and form current sheets while preserving connectivity.

15.3 How resistivity enables reconnection

At finite η,

$$\frac{D\psi}{Dt} = \eta\nabla^2\psi = -\eta J. \tag{15.3}$$

The value of ψ attached to a moving fluid element can change. In a thin current sheet, $|J| = |\nabla^2\psi|$ becomes large enough that ηJ remains dynamically important even if $\eta \ll 1$. Flux contours can then change connectivity near an X-point. This process is magnetic reconnection.

From Eq. (9.4), the out-of-plane electric field in the adopted gauge is $E_z = -\psi_t$ in dimensional variables. At a reconnection site, the non-ideal electric field is closely connected to the rate at which flux changes. Exact signs of a reported "reconnection rate" depend on which difference of O-point and X-point flux is defined, so Module 2 must record that diagnostic definition explicitly.

15.4 Current-sheet formation

The ideal terms can drive initially smooth fields toward smaller spatial scales. Since

$$J = -\nabla^2\psi, \tag{15.4}$$

strong curvature of the flux function produces large current. Diffusion acts increasingly strongly as the length scale shrinks. A useful order-of-magnitude balance across a sheet of thickness δ is

$$\eta\nabla^2\psi \sim \eta\frac{\psi}{\delta^2}. \tag{15.5}$$

Thus a small global diffusivity does not imply a negligible local resistive term.

15.5 Magnetic islands and coalescence

A magnetic island consists of closed flux contours around an O-point, bounded by a separatrix passing through one or more X-points. Parallel out-of-plane current channels can attract through the Lorentz force. Their magnetic islands are advected toward one another, a current sheet forms between them, reconnection changes the connectivity, and the islands merge into a larger structure. This is the *island-coalescence* process selected for Module 2.

The benchmark is valuable because it simultaneously exercises:

- nonlinear flux advection $[U, \psi]$;
- nonlinear vorticity advection $[U, \omega]$;
- Lorentz forcing $[J, \psi]$;
- current-sheet formation;
- resistive topology change;
- viscous damping; and
- parameter sensitivity to η, ν, and perturbation amplitude A_0.

The exact equilibrium and perturbation formulas, the number and placement of islands, and the operational definition of reconnected flux belong to Module 2. Module 1 establishes only the physics and conventions needed to interpret them.

15.6 A qualitative Sweet-Parker scale estimate

In a steady two-dimensional reconnection layer of length L_s and thickness δ_s, incompressible mass conservation suggests

$$V_{\mathrm{in}} L_s \sim V_{\mathrm{out}} \delta_s, \tag{15.6}$$

where V_{in} and V_{out} are inflow and outflow speeds. Balancing advective and resistive terms in the induction equation suggests

$$V_{\mathrm{in}} \frac{B}{\delta_s} \sim \eta_{\mathrm{m}} \frac{B}{\delta_s^2}, \qquad V_{\mathrm{in}} \sim \frac{\eta_{\mathrm{m}}}{\delta_s}. \tag{15.7}$$

If V_{out} is of order the Alfvén speed, these estimates give

$$\frac{\delta_s}{L_s} \sim S_s^{-1/2}, \qquad \frac{V_{\mathrm{in}}}{V_A} \sim S_s^{-1/2}, \tag{15.8}$$

where $S_s = L_s V_A / \eta_{\mathrm{m}}$ is based on sheet length. This is the classical Sweet-Parker scaling. It is an explanatory estimate, not an assumed law for every transient coalescence trajectory. At sufficiently large Lundquist number, secondary islands and unsteady dynamics may invalidate the single steady-sheet picture [2].

16 From equations to physics-informed residuals

16.1 What an equation residual means

A field is an exact solution of a partial differential equation (PDE) if substitution makes the left-minus-right difference zero everywhere in its domain. This difference is the *residual*. For predicted fields ψ_θ and U_θ, where θ denotes neural-network parameters, define

$$J_\theta = -\nabla_\perp^2 \psi_\theta, \tag{16.1}$$

$$\omega_\theta = \nabla_\perp^2 U_\theta. \tag{16.2}$$

The frozen *strong-form* residuals, meaning the differential equations evaluated point by point, are

$$\boxed{\mathcal{R}_\psi = \frac{\partial \psi_\theta}{\partial t} + [U_\theta, \psi_\theta] - \eta \nabla_\perp^2 \psi_\theta,} \tag{16.3}$$

$$\boxed{\mathcal{R}_\omega = \frac{\partial \omega_\theta}{\partial t} + [U_\theta, \omega_\theta] - [J_\theta, \psi_\theta] - \nu \nabla_\perp^2 \omega_\theta.} \tag{16.4}$$

An exact smooth solution has $\mathcal{R}_\psi = \mathcal{R}_\omega = 0$.

16.2 Residual losses do not replace data automatically

Using IC for *initial condition* and BC for *boundary condition*, a typical PINN objective contains

$$\mathcal{L} = w_{\mathrm{data}}\mathcal{L}_{\mathrm{data}} + w_{\mathrm{PDE}}\mathcal{L}_{\mathrm{PDE}} + w_{\mathrm{IC}}\mathcal{L}_{\mathrm{IC}} + w_{\mathrm{BC}}\mathcal{L}_{\mathrm{BC}} + w_{\mathrm{global}}\mathcal{L}_{\mathrm{global}}. \tag{16.5}$$

Here:

- $\mathcal{L}_{\mathrm{data}}$ compares predictions with sampled synthetic fields;
- $\mathcal{L}_{\mathrm{PDE}}$ penalizes $\mathcal{R}_\psi$ and $\mathcal{R}_\omega$ at collocation points, which are sampled space-time-parameter points where the residual is evaluated;
- $\mathcal{L}_{\mathrm{IC}}$ enforces the initial condition;
- $\mathcal{L}_{\mathrm{BC}}$ enforces periodic values and required derivatives;
- $\mathcal{L}_{\mathrm{global}}$ may penalize violations of energy, means, or other integral identities; and
- the nonnegative w values are loss weights.

The physics loss narrows the admissible function space, but the PDE alone does not select a unique trajectory without initial conditions, boundary conditions, parameters, and gauge constraints.

16.3 Derivative order in the direct (ψ, U) formulation

The direct residual is more demanding than its compact notation suggests. Substituting $J = -\nabla^2\psi$ and $\omega = \nabla^2 U$ into Eq. (16.4) produces:

- the mixed derivative $\partial_t \nabla^2 U$;
- third spatial derivatives of ψ inside $[J, \psi]$; and
- fourth spatial derivatives of U inside $\nabla^2\omega = \nabla^4 U$.

Automatic differentiation, an algorithm that differentiates the neural computational graph by repeated chain rules, can compute these derivatives. High-order derivatives can nevertheless be noisy to optimize and expensive to evaluate.

An alternative auxiliary-field formulation predicts (ψ, U, J, ω) and adds definition residuals

$$\mathcal{R}_J = J + \nabla^2\psi, \qquad \mathcal{R}_{\omega,\mathrm{def}} = \omega - \nabla^2 U. \tag{16.6}$$

Then the evolution residuals require at most first derivatives of J and second derivatives of ω. This changes the neural architecture and loss design, not the underlying physics. The version-1 project map remains

$$(x, y, t, \eta, \nu, A_0) \longmapsto (\psi, U) \tag{16.7}$$

unless a later implementation decision is explicitly versioned.

16.4 Periodic and gauge residuals

For direct coordinate inputs, periodic losses may include

$$\psi(0, y, t) - \psi(L_x, y, t) = 0, \tag{16.8}$$
$$U(0, y, t) - U(L_x, y, t) = 0, \tag{16.9}$$

and analogous y-face equations. Because the PDE contains high derivatives, matching relevant derivatives across the faces is also necessary unless periodicity is built into the network representation, for example through sine-cosine features.

The gauge constraint may be enforced by subtracting the spatial mean of U or by a loss on $\langle U \rangle$. The mean of ψ must remain consistent with the selected case metadata.

16.5 Global balance as a diagnostic or loss

For predicted fields, define the energy-balance defect

$$\mathcal{R}_E(t) = \frac{\mathrm{d}}{\mathrm{d}t}(E_B + E_K) + \eta \int_\Omega J^2 \,\mathrm{d}A + \nu \int_\Omega \omega^2 \,\mathrm{d}A. \tag{16.10}$$

An exact solution has $\mathcal{R}_E = 0$. A global loss can penalize sampled values of $\mathcal{R}_E$, but it cannot replace pointwise PDE residuals: many incorrect fields can share the same total energy history.

16.6 Parameter conditioning

The coefficients η, ν, and A_0 are inputs, not trainable physical constants in the initial parameterized surrogate. Whole parameter combinations must be held out for testing. If randomly selected space-time points from every trajectory are placed in both training and test sets, the test does not demonstrate generalization to an unseen physical case.

Important distinction

A small numerical PDE loss is not, by itself, proof of physical accuracy. Residual sampling may miss localized current sheets; a network may satisfy a modified equation because derivative scaling is wrong; and a low unweighted residual can hide an inaccurate high-order field such as J. Validation must include field errors, derived-field errors, global balances, and held-out parameter cases.

17 Relationship to M3D-C1

17.1 What M3D-C1 is

M3D-C1 is an initial-value, high-order finite-element code for nonlinear extended-MHD calculations in magnetized plasmas. The official overview describes models for density, fluid velocity, total and electron pressures, generalized Ohm's law, magnetic evolution, anisotropic transport, viscous stresses, sources, and optional two-fluid terms. It uses unstructured high-order C^1 elements and implicit or semi-implicit time integration to address disparate spatial and temporal scales [3, 4].

M3D-C1 also provides reduced representations based on scalar potentials. In cylindrical coordinates (R, φ, Z), where R is major-radius coordinate, φ is toroidal angle, and Z is vertical coordinate, its magnetic field can be represented schematically using a poloidal-flux variable ψ and a toroidal-field variable. A two-field reduced option retains flux-like and stream-function-like fields called ψ and U [3].

17.2 Conceptual connection without false equivalence

Our Cartesian potentials and the M3D-C1 reduced potentials share a concept: represent divergence-constrained vector fields using scalar functions and evolve a smaller field set. They are not numerically identical variables. M3D-C1 uses toroidal geometry, code-specific normalization, finite-element bases, and model options absent here. A later adapter must establish an explicit mapping from exported M3D-C1 quantities to any learning variables.

Feature	Version-1 synthetic model	M3D-C1 capability or context
Geometry	2D Cartesian periodic square	Cylindrical/toroidal geometry; axisymmetric and fully 3D modes
Primary dynamics	Two scalar fields ψ, U	Single-fluid, two-fluid, and reduced model options with more fields
Density	Constant and not evolved	Evolved density in general models
Pressure/temperature	Not evolved	Total/electron pressure and thermal transport options

Feature	Version-1 synthetic model	M3D-C1 capability or context
Magnetic field	Uniform guide field plus in-plane flux field	General toroidal magnetic representation
Flow	In-plane incompressible flow	More general velocity decomposition and compressible dynamics
Non-ideal physics	Constant scalar η and ν	Resistivity, stress models, two-fluid terms, transport, sources, and additional closures
Boundary	Doubly periodic	Plasma, vacuum, wall, and free-boundary capabilities depending on problem
Spatial method	Planned Fourier pseudo-spectral solver	High-order unstructured C^1 finite elements
Time advance	Selected and verified in Module 2	Implicit and semi-implicit methods
Data status	Independently generated synthetic fields	Genuine code exports require licensed runs or provided data
Scientific claim	Reduced-MHD PINN proof of concept	No validated M3D-C1 surrogate claim without genuine data

17.3 What the proof of concept legitimately demonstrates

If successful, the project can show that a parameterized neural surrogate can learn nonlinear resistive-MHD field evolution while being constrained by a stated PDE, initial conditions, periodic boundaries, and global balances. It can also establish versioned metadata, held-out case splits, diagnostics, and an export schema suitable for later replacement of synthetic trajectories by genuine M3D-C1 results.

It cannot establish that the neural network predicts toroidal extended-MHD behavior, pressure-driven modes, two-fluid dynamics, stellarator geometry, wall interaction, or M3D-C1 numerical errors. Those are separate scientific questions.

18 Worked analytic examples

These examples are intentionally simple. Their purpose is to turn the definitions into calculable objects and to provide future unit or manufactured-solution tests.

18.1 Example 1: a single magnetic Fourier mode

Let

$$\psi(x, y) = A\cos(k_x x)\cos(k_y y), \qquad U = 0, \tag{18.1}$$

where A is a constant amplitude and k_x, k_y are wavenumbers compatible with the periodic box. Then

$$B_x = \psi_y = -Ak_y\cos(k_x x)\sin(k_y y), \tag{18.2}$$

$$B_y = -\psi_x = +Ak_x\sin(k_x x)\cos(k_y y), \tag{18.3}$$

$$J = -\nabla^2\psi = (k_x^2 + k_y^2)\psi. \tag{18.4}$$

Because J is proportional to ψ,

$$[J, \psi] = 0. \tag{18.5}$$

There is no vorticity generation. The flux equation reduces to diffusion:

$$\psi_t = -\eta(k_x^2 + k_y^2)\psi. \tag{18.6}$$

The exact solution is

$$\boxed{\psi(x, y, t) = A\exp[-\eta(k_x^2 + k_y^2)t]\cos(k_x x)\cos(k_y y), \qquad U = 0.} \tag{18.7}$$

This solution checks spectral derivatives, current sign, resistive decay rate, and time integration.

18.2 Example 2: a single viscously decaying flow mode

Let

$$U(x, y, t) = C(t)\cos(k_x x)\cos(k_y y), \qquad \psi = 0. \tag{18.8}$$

Then

$$v_x = C(t)k_y\cos(k_x x)\sin(k_y y), \tag{18.9}$$

$$v_y = -C(t)k_x\sin(k_x x)\cos(k_y y), \tag{18.10}$$

$$\omega = -(k_x^2 + k_y^2)U. \tag{18.11}$$

Because ω is proportional to U, $[U, \omega] = 0$. The vorticity equation becomes

$$\omega_t = \nu\nabla^2\omega, \tag{18.12}$$

which implies

$$\boxed{C(t) = C(0)\exp[-\nu(k_x^2 + k_y^2)t].} \tag{18.13}$$

This checks stream-function signs, vorticity, Poisson inversion, and viscous decay.

18.3 Example 3: the bracket as advection

Choose

$$U = \sin x \sin y, \qquad f = \cos x. \tag{18.14}$$

Then

$$\boldsymbol{v} = (-U_y, U_x) = (-\sin x \cos y, \cos x \sin y), \tag{18.15}$$

$$\boldsymbol{v} \cdot \nabla f = (-\sin x \cos y)(-\sin x) + (\cos x \sin y)(0) \tag{18.16}$$

$$= \sin^2 x \cos y. \tag{18.17}$$

Directly from the bracket,

$$[U, f] = U_x f_y - U_y f_x \tag{18.18}$$

$$= (\cos x \sin y)(0) - (\sin x \cos y)(-\sin x) \tag{18.19}$$

$$= \sin^2 x \cos y. \tag{18.20}$$

The results match, verifying Eq. (8.4).

18.4 Example 4: an ideal static equilibrium family

Let $U = 0$ and suppose

$$J = F(\psi) \tag{18.21}$$

for any differentiable single-valued function F. Then

$$[J, \psi] = F'(\psi)[\psi, \psi] = 0. \tag{18.22}$$

With $\eta = \nu = 0$, both evolution equations are satisfied. This is the curl-level equilibrium condition. The pressure field removed by the curl may still be needed to balance the gradient part of the Lorentz force in the original momentum equation.

18.5 Example 5: energy decay of the magnetic mode

For the exact solution in Eq. (18.7), magnetic energy is proportional to the squared amplitude:

$$E_B(t) = E_B(0) \exp[-2\eta(k_x^2 + k_y^2)t]. \tag{18.23}$$

The energy law predicts

$$\frac{\mathrm{d}E_B}{\mathrm{d}t} = -\eta \int J^2 \,\mathrm{d}A. \tag{18.24}$$

Since $J = k^2\psi$ and $\int |\nabla\psi|^2 \mathrm{d}A = k^2 \int \psi^2 \mathrm{d}A$, the right-hand side equals $-2\eta k^2 E_B$, exactly matching the time derivative. Here $k^2 = k_x^2 + k_y^2$.

19 Verification identities and mandatory convention tests

The following checks should be implemented before island-coalescence data is accepted as a learning target.

19.1 Differential identity tests

For known Fourier modes, verify:

1. first and second spatial derivatives at their expected convergence rate;
2. $\nabla \cdot \boldsymbol{B}_\perp = 0$ and $\nabla \cdot \boldsymbol{v}_\perp = 0$ near numerical precision;
3. $J = -\nabla^2\psi$ and $\omega = +\nabla^2 U$;
4. bracket antisymmetry, $[f, g] + [g, f] = 0$;
5. self-bracket, $[f, f] = 0$;
6. the cyclic integral identity in Eq. (8.13); and
7. periodic matching of fields and derivatives.

19.2 Analytic evolution tests

Use Eqs. (18.7) and its viscous counterpart to verify:

- exponential amplitude decay rate;
- correct independence of magnetic decay from ν in the no-flow case;
- correct independence of flow decay from η in the no-field case;
- convergence under grid refinement;
- convergence under time-step refinement; and
- preservation of all inactive fields at roundoff or truncation level.

19.3 Integral-balance tests

Track

$$E_B(t), \quad E_K(t), \quad \int J^2 \mathrm{d}A, \quad \int \omega^2 \mathrm{d}A, \quad \langle \psi \rangle, \quad \langle U \rangle, \quad \langle J \rangle, \quad \langle \omega \rangle. \tag{19.1}$$

The discrete energy defect over a time interval $[t_a, t_b]$ can be evaluated as

$$\epsilon_E = E(t_b) - E(t_a) + \int_{t_a}^{t_b} \left[\eta \int_\Omega J^2 \mathrm{d}A + \nu \int_\Omega \omega^2 \mathrm{d}A \right] \mathrm{d}t. \tag{19.2}$$

For a convergent consistent solver, ϵ_E should approach zero as spatial and temporal resolution increase.

19.4 Why visual plausibility is insufficient

Contour plots can look physically reasonable even when a current sign is reversed, a derivative is aliased, or energy grows spuriously. Verification must use exact identities and convergence, not only images. Conversely, a thin current sheet that appears noisy may be under-resolved rather than physically turbulent. Resolution studies distinguish these possibilities.

Check	Expected result	Likely failure indicated
$\nabla \cdot \boldsymbol{B}$	Near numerical precision	Flux-to-field derivative inconsistency
$\nabla \cdot \boldsymbol{v}$	Near numerical precision	Stream-to-velocity derivative inconsistency
$[f, g] + [g, f]$	Near zero	Bracket implementation or axis-order error
$\langle U \rangle$	Exactly or nearly zero	Uncontrolled Poisson zero mode
$\langle \psi \rangle$	Constant	Mean-mode drift or inconsistent gauge handling
$E(t)$ for $\eta, \nu > 0$	Nonincreasing	Sign error, forcing, or numerical instability
Analytic Fourier decay	Correct exponent	Diffusion sign, wavenumber, or time-scale error
Refinement study	Error decreases	Under-resolution or inconsistent discretization

20 Common misconceptions and sign traps

20.1 “ψ is the magnetic field”

No. ψ is a scalar potential. The in-plane magnetic field is made from its first derivatives. A spatially constant ψ produces zero in-plane magnetic field.

20.2 “U is a velocity component”

No. U is a stream function with velocity obtained from first derivatives. Adding a spatial constant to U changes no physical velocity.

20.3 “J is the current density in amperes per square meter”

Not in the nondimensional project. Physical current density is

$$j_{z,\mathrm{d}} = \frac{B_0}{\mu_0 L_0} J. \tag{20.1}$$

The project J is dimensionless.

20.4 “Resistivity and magnetic diffusivity have the same units”

No. Electrical resistivity ρ_Ω has units $\Omega\,\mathrm{m}$. Magnetic diffusivity $\eta_\mathrm{m} = \rho_\Omega/\mu_0$ has units $\mathrm{m}^2\,\mathrm{s}^{-1}$. Project η is dimensionless.

20.5 “A small η can be ignored everywhere”

No. The relevant term is $\eta\nabla^2\psi = -\eta J$. Current density can become large in a thin sheet, making the product finite and controlling topology change.

20.6 “Incompressible means motionless or uniform”

No. It means $\nabla \cdot \boldsymbol{v} = 0$. Incompressible flows can be nonlinear, vortical, and turbulent.

20.7 “Two-dimensional means every vector has two components”

No. Two-dimensional means fields are independent of one coordinate. Current and vorticity point in the ignorable z direction even though magnetic and velocity perturbations lie in the plane.

20.8 “Pressure was set to zero”

No. Its gradient was eliminated by taking the curl under constant density. Pressure still enforces incompressibility in the parent momentum equation.

20.9 “Reduced MHD and full M3D-C1 are the same equations”

No. M3D-C1 supports far richer geometry, field content, transport, and closures. Shared variable names do not imply numerical identity.

20.10 “A PINN residual is just another regression error”

Not exactly. It measures violation of a differential equation. Its magnitude depends on derivative scales, normalization, and sampling. A residual can be small for the wrong equation if chain-rule factors or signs are incorrect.

20.11 “Numerical diffusion and physical diffusion are interchangeable”

No. The parameters η and ν represent explicit model transport. Discretization also introduces truncation error and may introduce numerical dissipation. The latter must be shown to decrease with refinement and should not silently replace the specified physics.

21 Module-1 computational contract

The following statements constitute the handoff from physics notes to later implementation.

21.1 State and parameter contract

$$\text{Inputs: } (x, y, t, \eta, \nu, A_0), \tag{21.1}$$

$$\text{Primary outputs: } (\psi, U), \tag{21.2}$$

$$\text{Derived outputs: } J = -\nabla^2\psi, \quad \omega = \nabla^2 U. \tag{21.3}$$

All are dimensionless in version 1. Dimensional reference scales must remain in metadata even when assigned numerical value one.

21.2 Equation contract

$$\psi_t + [U, \psi] - \eta \nabla^2 \psi = 0, \tag{21.4}$$

$$\omega_t + [U, \omega] - [J, \psi] - \nu \nabla^2 \omega = 0, \tag{21.5}$$

$$[f, g] = f_x g_y - f_y g_x. \tag{21.6}$$

21.3 Boundary and gauge contract

- ψ, U, and required derivatives are periodic in x and y.
- $\langle U \rangle = 0$ at every stored time.
- $\langle \psi \rangle$ is recorded and remains constant within a case.
- The $\boldsymbol{k} = 0$ mode is handled explicitly in every Poisson inversion.

21.4 Diagnostic contract

At minimum, store or compute:

- magnetic energy E_B;
- kinetic energy E_K;
- total energy $E_B + E_K$;
- peak current $\max_\Omega |J|$;
- Ohmic dissipation rate $\eta \int J^2 \mathrm{d}A$;
- viscous dissipation rate $\nu \int \omega^2 \mathrm{d}A$;
- mean and gauge checks;
- reconnected flux using a versioned O-point/X-point definition; and
- PDE residuals for learned fields.

21.5 Required provenance

Each generated case must preserve the domain, grid, time array, coefficients, initial-condition parameters, normalization scales, sign-convention version, solver version, boundary conditions, field names, and diagnostic definitions. A contour image is not a scientific dataset.

22 Summary and bridge to Module 2

The project begins from a conducting one-fluid description and then imposes constant density, incompressibility, two-dimensional Cartesian geometry, a uniform guide field, scalar resistivity and viscosity, and periodic boundaries. Divergence-free in-plane fields are represented by a flux function and stream function:

$$\boldsymbol{B}_\perp = (\psi_y, -\psi_x), \qquad \boldsymbol{v}_\perp = (-U_y, U_x). \tag{22.1}$$

These choices fix

$$J = -\nabla^2\psi, \qquad \omega = +\nabla^2 U. \tag{22.2}$$

Two-dimensional advection is represented by

$$[f, g] = f_x g_y - f_y g_x. \tag{22.3}$$

The final dimensionless system is

$$\psi_t + [U, \psi] = \eta\nabla^2\psi, \tag{22.4}$$

$$\omega_t + [U, \omega] = [J, \psi] + \nu\nabla^2\omega. \tag{22.5}$$

The ideal nonlinear terms exchange magnetic and kinetic energy but do not change their sum. On the periodic domain,

$$\frac{\mathrm{d}}{\mathrm{d}t}(E_B + E_K) = -\eta\int J^2 \mathrm{d}A - \nu\int \omega^2 \mathrm{d}A. \tag{22.6}$$

Finite resistivity breaks frozen-in flux in regions of high current, enabling reconnection and magnetic-island coalescence.

Module 2 must now supply what Module 1 intentionally does not: a precise island-coalescence equilibrium and perturbation; pseudo-spectral spatial discretization; de-aliasing (removal of spurious unresolved Fourier-mode interactions); time integration; case parameter ranges; convergence criteria; and a versioned definition of reconnected flux. No synthetic trajectory should be used for neural training until the analytic decay tests, sign audits, periodic checks, convergence studies, and energy balance described here have passed.

A Component-by-component sign audit

This appendix repeats the shortest possible audit in one place.

A.1 Magnetic field and current

$$\begin{aligned}
\boldsymbol{B}_\perp &= \nabla\psi \times \widehat{\boldsymbol{z}} && \text{(A.1)}\\
&= (\psi_x\widehat{\boldsymbol{x}} + \psi_y\widehat{\boldsymbol{y}}) \times \widehat{\boldsymbol{z}} && \text{(A.2)}\\
&= \psi_y\widehat{\boldsymbol{x}} - \psi_x\widehat{\boldsymbol{y}}, && \text{(A.3)}\\
(\nabla \times \boldsymbol{B})_z &= \partial_x(-\psi_x) - \partial_y(\psi_y) && \text{(A.4)}\\
&= -\nabla^2\psi. && \text{(A.5)}
\end{aligned}$$

Therefore $J = -\nabla^2\psi$.

A.2 Velocity and vorticity

$$\begin{aligned}
\boldsymbol{v}_\perp &= \widehat{\boldsymbol{z}} \times \nabla U && \text{(A.6)}\\
&= -U_y\widehat{\boldsymbol{x}} + U_x\widehat{\boldsymbol{y}}, && \text{(A.7)}\\
(\nabla \times \boldsymbol{v})_z &= \partial_x(U_x) - \partial_y(-U_y) && \text{(A.8)}\\
&= +\nabla^2 U. && \text{(A.9)}
\end{aligned}$$

Therefore $\omega = +\nabla^2 U$.

A.3 Advection

$$\boldsymbol{v}\cdot\nabla f = (-U_y)f_x + (U_x)f_y = [U, f]. \tag{A.10}$$

A.4 Flux equation

$$\begin{aligned}
(\boldsymbol{v}\times\boldsymbol{B})_z &= v_xB_y - v_yB_x && \text{(A.11)}\\
&= (-U_y)(-\psi_x) - (U_x)(\psi_y) && \text{(A.12)}\\
&= -[U,\psi], && \text{(A.13)}\\
E_z + (\boldsymbol{v}\times\boldsymbol{B})_z &= \rho_\Omega j_z, && \text{(A.14)}\\
-\psi_t - [U,\psi] &= -\eta_{\mathrm{m}}\nabla^2\psi, && \text{(A.15)}\\
\psi_t + [U,\psi] &= +\eta_{\mathrm{m}}\nabla^2\psi. && \text{(A.16)}
\end{aligned}$$

A.5 Lorentz bracket

$$\boldsymbol{j} \times \boldsymbol{B} = j_z \widehat{\boldsymbol{z}} \times (\psi_y, -\psi_x, 0) \tag{A.17}$$
$$= j_z(\psi_x, \psi_y, 0) \tag{A.18}$$
$$= j_z \nabla \psi, \tag{A.19}$$
$$[\nabla \times (j_z \nabla \psi)]_z = (j_z)_x \psi_y - (j_z)_y \psi_x \tag{A.20}$$
$$= [j_z, \psi]. \tag{A.21}$$

Alfvén normalization maps the physical current density to J and produces $+[J, \psi]$ in the vorticity equation.

B Vector and integral identities used in the energy proof

For smooth periodic scalar fields f and g,

$$\int_\Omega f \nabla^2 g \,\mathrm{d}A = -\int_\Omega \nabla f \cdot \nabla g \,\mathrm{d}A \tag{B.1}$$
$$= \int_\Omega g \nabla^2 f \,\mathrm{d}A. \tag{B.2}$$

Thus the periodic Laplacian is self-adjoint.

For smooth periodic f, g, h,

$$\int_\Omega f[g, h] \,\mathrm{d}A = \int_\Omega f(g_x h_y - g_y h_x) \,\mathrm{d}A \tag{B.3}$$
$$= -\int_\Omega g \left(f_x h_y - f_y h_x\right) \mathrm{d}A \tag{B.4}$$
$$= \int_\Omega g[h, f] \,\mathrm{d}A. \tag{B.5}$$

The omitted boundary terms vanish by periodicity. Repeating the permutation gives the full cyclic identity.

C Symbol table

Symbol	Meaning	Units or status
x, y, z	Cartesian coordinates after physical nondimensionalization	dimensionless
$x_\mathrm{d}, y_\mathrm{d}, z_\mathrm{d}$	Dimensional Cartesian coordinates	m
t, t_d	Dimensionless and dimensional time	1 and s

Symbol	Meaning	Units or status
Ω	Periodic two-dimensional domain	dimensionless area
L_x, L_y	Dimensionless domain lengths	dimensionless
L_0	Dimensional reference length	m
$\widehat{\boldsymbol{x}}, \widehat{\boldsymbol{y}}, \widehat{\boldsymbol{z}}$	Cartesian unit vectors	dimensionless
∇, ∇^2	Gradient and Laplacian in dimensionless coordinates	dimensionless operators
∂_t	Partial derivative at fixed spatial position	inverse dimensionless time
D/Dt	Material derivative following the fluid	inverse time
$[f, g]$	Poisson bracket $f_x g_y - f_y g_x$	depends on f, g
ρ_{d}	Mass density	$\mathrm{kg\,m^{-3}}$
ρ_0	Constant reference mass density	$\mathrm{kg\,m^{-3}}$
n_i, n_e	Ion and electron number density	$\mathrm{m^{-3}}$
m_i, m_e	Ion and electron mass	kg
e	Positive elementary-charge magnitude	C
p_{d}	Isotropic fluid pressure	Pa
γ	Ratio of specific heats	dimensionless
$\boldsymbol{E}_{\mathrm{d}}$	Electric field	$\mathrm{V\,m^{-1}}$
$\boldsymbol{B}_{\mathrm{d}}$	Magnetic field	T
B_g	Constant out-of-plane guide field	T
B_0	Reference in-plane magnetic-field scale	T
$\boldsymbol{j}_{\mathrm{d}}$	Electric current density	$\mathrm{A\,m^{-2}}$
$j_{z,\mathrm{d}}$	Out-of-plane current-density component	$\mathrm{A\,m^{-2}}$
μ_0	Permeability of free space	$\mathrm{N\,A^{-2}}$
ρ_Ω	Electrical resistivity	$\Omega\,\mathrm{m}$
η_{m}	Magnetic diffusivity ρ_Ω/μ_0	$\mathrm{m^2\,s^{-1}}$
η	Project magnetic diffusivity $\eta_{\mathrm{m}}/(L_0 V_A)$	dimensionless
ν_{d}	Dimensional kinematic viscosity	$\mathrm{m^2\,s^{-1}}$
ν	Project kinematic viscosity $\nu_{\mathrm{d}}/(L_0 V_A)$	dimensionless
V_A	Alfvén speed $B_0/\sqrt{\mu_0\rho_0}$	$\mathrm{m\,s^{-1}}$
t_A	Alfvén time L_0/V_A	s
S	Lundquist number $L_0 V_A/\eta_{\mathrm{m}}$	dimensionless
Re	Reynolds number $L_0 V_A/\nu_{\mathrm{d}}$	dimensionless
Rm	Magnetic Reynolds number $L_0 V_0/\eta_{\mathrm{m}}$	dimensionless
Pm	Magnetic Prandtl number $\nu_{\mathrm{d}}/\eta_{\mathrm{m}}$	dimensionless
ψ_{d}	Dimensional magnetic-flux function	T m
ψ	Dimensionless magnetic-flux function	dimensionless
U_{d}	Dimensional flow stream function	$\mathrm{m^2\,s^{-1}}$
U	Dimensionless flow stream function	dimensionless
J	Dimensionless current variable $-\nabla^2\psi$	dimensionless
ω	Dimensionless out-of-plane vorticity $\nabla^2 U$	dimensionless
E_B, E_K	Dimensionless magnetic and kinetic energy integrals	dimensionless
H_C	Dimensionless cross helicity	dimensionless
A_0	Initial perturbation amplitude parameter	dimensionless

Symbol	Meaning	Units or status
$\boldsymbol{\mu}$	Case parameter vector (η, ν, A_0)	dimensionless
θ	Neural-network trainable parameter vector	numerical parameters
$\mathcal{R}_\psi, \mathcal{R}_\omega$	Flux and vorticity PDE residuals	dimensionless residuals
$\langle f \rangle$	Spatial mean of f over Ω	same units as f

D Acronym and terminology table

Term	Meaning
1D, 2D, 3D	One-, two-, and three-dimensional
BC	Boundary condition
C^1	Continuity class in which a represented field and its first derivatives are continuous
FFT	Fast Fourier transform
IC	Initial condition
M3D-C1	Code name for a high-order extended-MHD simulation framework; C1 refers to C^1 elements
MHD	Magnetohydrodynamics
O-point	Magnetic critical point at the center of nested closed flux contours
PDE	Partial differential equation
PINN	Physics-informed neural network
RMHD	Reduced magnetohydrodynamics; the exact reduction must always be stated
SI	International System of Units
X-point	Saddle-type magnetic critical point on a separatrix
Alfvén speed	Characteristic MHD wave speed $B/\sqrt{\mu_0 \rho}$
Advection	Transport by bulk fluid motion
Diffusion	Smoothing driven by spatial curvature and a diffusivity
Flux freezing	Ideal-MHD preservation of magnetic flux attached to moving fluid elements
Guide field	Strong magnetic-field component in the ignorable direction
Magnetic island	Region of closed flux contours surrounding an O-point
Magnetic reconnection	Non-ideal change of magnetic-field-line connectivity
Material derivative	Time rate of change following a moving fluid element
Separatrix	Flux contour separating topologically distinct field-line regions

Term	Meaning
Stream function	Scalar potential whose rotated gradient produces incompressible 2D velocity
Vorticity	Curl of velocity; local rotational content of the flow

E Concept questions and answer sketches

Questions

Q1. Why is MHD a fluid theory even though its derivation begins from kinetic equations?

Q2. What is the physical difference between bulk velocity and electric current density?

Q3. Why does writing $\boldsymbol{B}_\perp = \nabla\psi \times \hat{\boldsymbol{z}}$ guarantee $\nabla \cdot \boldsymbol{B} = 0$?

Q4. With the frozen convention, why does current have a minus sign but vorticity a plus sign?

Q5. Show directly that $[U, f] = \boldsymbol{v} \cdot \nabla f$.

Q6. What happens to ψ following a fluid element when $\eta = 0$?

Q7. Why can a small resistivity control the evolution inside a current sheet?

Q8. Why does pressure disappear from the vorticity equation?

Q9. What does the term $[J, \psi]$ represent physically?

Q10. Why is the mean of U a gauge choice?

Q11. Why must the zero Fourier mode be treated separately when solving $\nabla^2 U = \omega$?

Q12. Which nonlinear term transfers energy between magnetic and kinetic reservoirs?

Q13. Why can total energy only decrease for $\eta, \nu > 0$ in the unforced periodic model?

Q14. Why is neural min-max scaling not the same as physical nondimensionalization?

Q15. Why would a random point split across every trajectory overstate surrogate generalization?

Q16. In what precise sense is this project connected to M3D-C1, and in what sense is it not?

Answer sketches

A1. MHD evolves a finite set of spatial moments such as density, bulk velocity, pressure, and magnetic field, rather than the full velocity-space distribution functions.

A2. Bulk velocity is mass-weighted common motion. Current density is charge-weighted relative motion of positive and negative species.

A3. It is a curl: $\boldsymbol{B}_\perp = \nabla \times (\psi \hat{\boldsymbol{z}})$, and the divergence of any smooth curl is zero.

A4. The magnetic and velocity potentials use opposite cross-product order. Taking their z curls gives $J = -\nabla^2\psi$ and $\omega = +\nabla^2 U$.

A5. Substitute $\boldsymbol{v} = (-U_y, U_x)$: $\boldsymbol{v} \cdot \nabla f = -U_y f_x + U_x f_y = [U, f]$.

A6. Its material derivative is zero; the fluid element retains its flux label, expressing frozen-in topology.

A7. The resistive term is $-\eta J$. A current sheet has very large J because ψ varies over a small thickness.

A8. Under constant density, pressure contributes a pure gradient to acceleration, and curl of a gradient is zero.

A9. It is the out-of-plane curl of the Lorentz force per unit mass and drives vorticity and magnetic-kinetic energy exchange.

A10. Only spatial derivatives of U affect velocity and vorticity. Adding a spatial constant changes neither.

A11. The Laplacian eigenvalue is $-|\boldsymbol{k}|^2$, which is zero at $\boldsymbol{k} = 0$; division is undefined. The gauge sets the mean of U.

A12. The domain integral $\int U[J, \psi]\mathrm{d}A$ appears with opposite signs in magnetic and kinetic energy budgets.

A13. Periodic boundaries remove surface fluxes, the brackets only redistribute, and the remaining sinks are nonnegative $\eta \int J^2$ and $\nu \int \omega^2$.

A14. Physical nondimensionalization defines the PDE coefficients. Neural scaling is an additional coordinate transformation whose chain-rule factors must be included in derivatives.

A15. The model can interpolate within portions of every already-seen trajectory. Holding out whole parameter combinations tests a genuinely unseen physical case.

A16. It demonstrates a two-field reduced-MHD surrogate workflow and data contract conceptually relevant to reduced MHD. It neither runs nor validates the full toroidal extended-MHD code.

References

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Addendum: full-field operator surrogate formulation

A.1 Purpose and status of this addendum

The reduced-MHD physics developed in this module is unchanged. The governing fields, sign conventions, normalization, periodic domain, Poisson bracket, current and vorticity definitions, and evolution equations remain the physical foundation of the project. The update considered here changes the *surrogate-learning map*: instead of treating each space–time coordinate as an independent neural input, the surrogate may consume an entire initial field and learn an operator that maps that field to a later field.

The earlier coordinate-network formulation remains a valid baseline,

$$(x, y, t, \eta, \nu, A_0) \longmapsto (\psi, U), \tag{A.1}$$

but the proposed full-field formulation is

$$\boxed{(\psi_0(x, y), U_0(x, y), \eta, \nu, t) \longmapsto (\psi(x, y, t), U(x, y, t)).} \tag{A.2}$$

Equation (A.2) is a *field-to-field operator-learning problem.* The network input and output are numerical field arrays, not rendered contour plots, screenshots, or RGB images. Preserving the raw arrays retains sign, normalization, spatial resolution, and derivative information required by the MHD equations.

A.2 What happens to the perturbation amplitude A_0?

For the version-1 island-coalescence initial condition, Module 2 specifies

$$U_0(x, y) = A_0(\cos x - \cos y). \tag{A.3}$$

In the coordinate-network formulation, A_0 must be supplied explicitly because the network receives only the scalar coordinates and case parameters. In the full-field formulation, the magnitude and spatial structure of the perturbation are already encoded in the input field $U_0(x, y)$. Therefore A_0 need not be a separate neural input when U_0 itself is provided.

This distinction is useful:

Formulation	**How the initial perturbation enters**
Coordinate surrogate	Scalar A_0 identifies the amplitude of a prescribed perturbation shape.
Full-field operator	The complete array $U_0(x, y)$ carries both perturbation amplitude and spatial shape.

The operator formulation therefore creates a path beyond a one-parameter family of perturbations. Future datasets may contain different admissible initial fields ψ_0 and U_0, provided each case is generated and verified under a clearly versioned physical contract. The network could then learn responses to changes in the *shape* of the initial state, not merely to a change in the scalar amplitude A_0.

A.3 Physical interpretation of the operator map

A useful way to distinguish the two learning problems is

$$\text{coordinate model:} \quad (\text{where}, \text{when}, \text{which scalar case}) \longmapsto \text{field value at one point}, \tag{A.4}$$

$$\text{operator model:} \quad (\text{entire initial plasma state}, \text{case coefficients}, \text{time}) \longmapsto \text{entire later plasma state}. \tag{A.5}$$

The latter is closer to the way a time-dependent simulation is naturally viewed: an initial state is supplied, the equations evolve that state, and a later field state is produced. For the present two-field reduced model, the minimal operator input can be (ψ_0, U_0) together with the transport coefficients η and ν and a requested time t. The output remains (ψ, U); the derived fields are reconstructed from

$$J = -\nabla_\perp^2 \psi, \qquad \omega = \nabla_\perp^2 U. \tag{A.6}$$

The physical equations that can later constrain a physics-informed operator are exactly the same reduced-MHD equations frozen earlier in this module:

$$R_\psi = \partial_t \psi + [U, \psi] - \eta \nabla_\perp^2 \psi, \tag{A.7}$$

$$R_\omega = \partial_t \omega + [U, \omega] - [J, \psi] - \nu \nabla_\perp^2 \omega. \tag{A.8}$$

A data-only operator can first be trained against reference trajectories. A physics-informed version can then penalize Eqs. (A.7)–(A.8), together with periodicity, definitions, initial-state consistency, and other structural constraints. Thus the architectural change does not alter the MHD model; it changes the mathematical object approximated by the neural network.

A.4 Consequences for the downstream modules

If the full-field operator formulation is adopted, the downstream workflow changes as follows:

Module	**Required change**
1	Small. Preserve the reduced-MHD derivation and conventions; append the operator-learning objective.
2	Moderate. Preserve the verified solver, but expose trajectories explicitly as initial-field-to-later-field training pairs and eventually permit versioned families of initial fields.
3	Large. Redefine dataset samples, tensor shapes, batching, normalization, and whole-case splits around full field arrays rather than pointwise coordinate rows.
4	Large. Replace the coordinate MLP baseline with a data-only neural operator baseline, for example a Fourier neural operator or another field-to-field architecture.
5	Large. Apply reduced-MHD residuals and structural constraints to the operator predictions rather than to a coordinate MLP.

A.5 Relationship to M3D-C1 and claim boundary

The operator formulation can make the *data interface* look more like the natural output of a large simulation code because both are organized around spatial fields. That similarity must not be confused with physical fidelity. Fields generated by the Module 2 periodic Cartesian two-field solver remain *synthetic reduced-MHD data*. They are not M3D-C1 calculations, even if the arrays are stored using names, shapes, metadata, or interfaces chosen to facilitate later transfer.

Accordingly, the scientifically defensible progression is

$$\begin{aligned} \text{verified reduced-MHD trajectories} &\longrightarrow \text{data-only field operator} \\ &\longrightarrow \text{physics-informed field operator} \\ &\longrightarrow \text{possible later transfer to genuine higher-fidelity data.} \end{aligned} \tag{A.9}$$

The first three stages demonstrate a surrogate methodology for nonlinear resistive reduced MHD. A claim about M3D-C1 itself would require genuine M3D-C1 outputs and a separate validation/transfer study.

A.6 Updated learning objective

The proposed operator-learning objective for the reduced proof of concept is:

> Given a verified initial reduced-MHD field state (ψ_0, U_0), transport coefficients (η, ν), and a requested time t, learn the nonlinear solution operator that predicts the corresponding full fields (ψ, U), while retaining the option to enforce the reduced-MHD equations and structural constraints during training.

This objective is broader than learning the response to the single scalar parameter A_0, because the input can eventually represent a family of spatially distinct initial conditions. For the first implementation, however, the existing island-coalescence family remains an appropriate controlled benchmark: it allows the operator pipeline to be verified before increasing the diversity of initial fields.

A.7 Addendum change note

This addendum was appended on 11 August 2026. No earlier section or equation of Module 1 was deleted or rewritten. The added material records a proposed transition from the pointwise coordinate surrogate to a full-field operator surrogate, explains how A_0 becomes encoded in U_0, states which reduced-MHD physics remains unchanged, and records the expected impact on Modules 2–5.